Computer Networks

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Objectives

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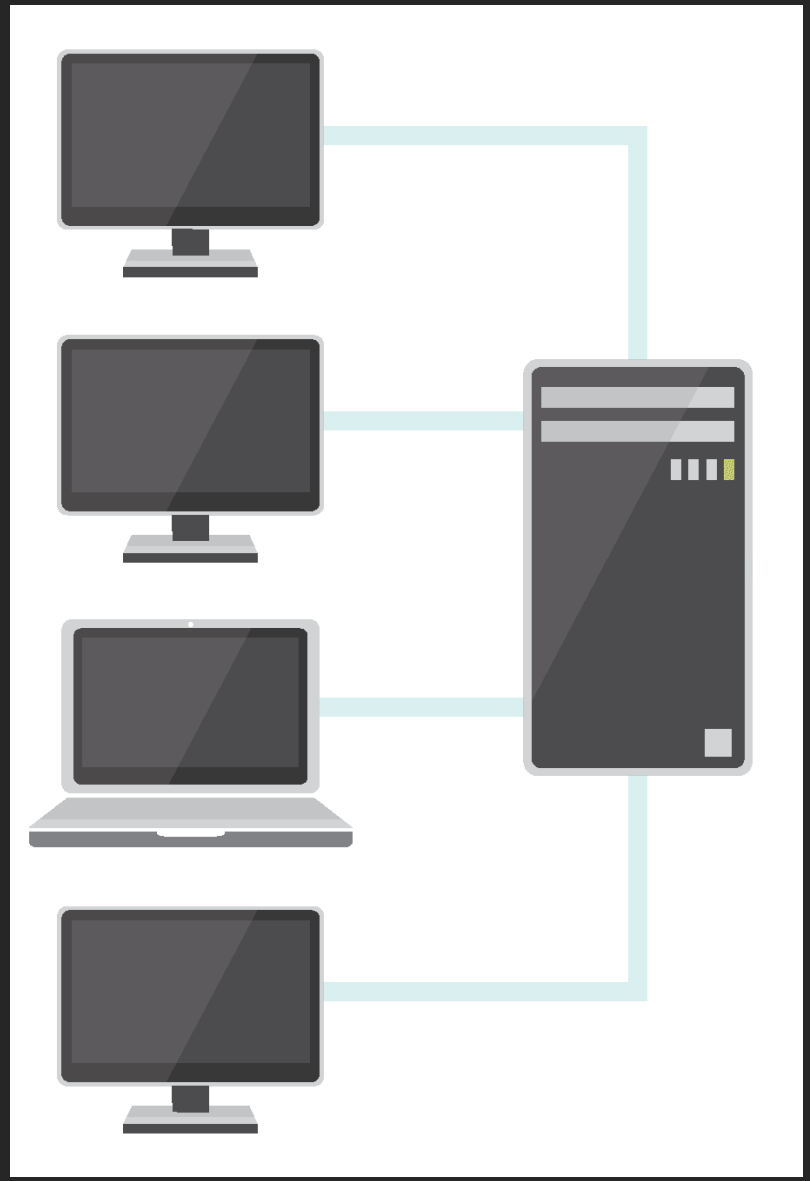
1. Network Types

Networks

- Two or more computers connected together in a way that allows resource sharing.
  - A network contains any combination of computers, computer terminals, printers, display devices, cables, or wireless connections.
  - A network is a collection of computers or other hardware devices that are connected together using special hardware and software. This allows them to share information and cooperate.

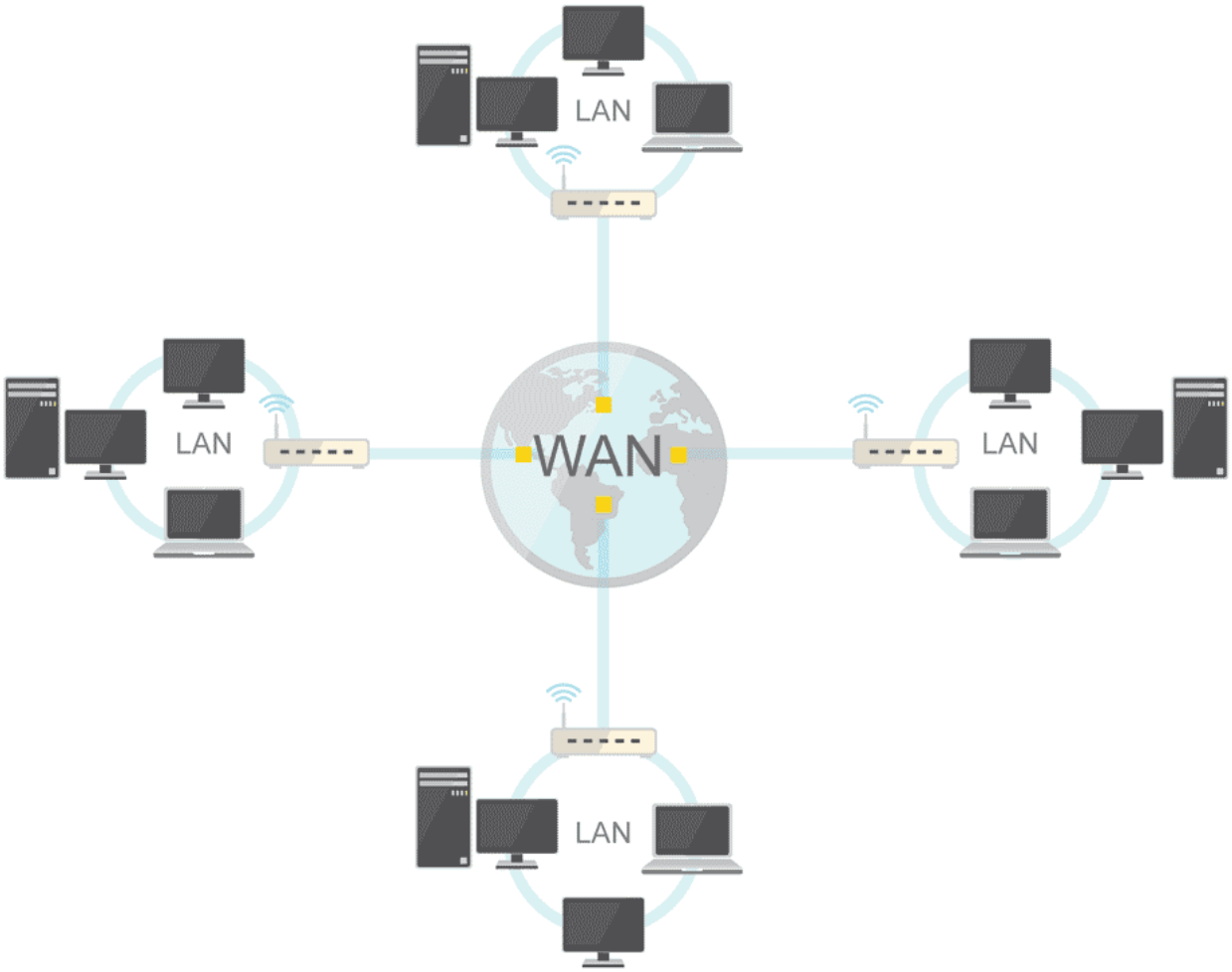
Network types

- Computer networks vary in shape (topology) and size depending on their application.
- Some of the major types are:
  - Local area networks (LANs).
  - Wide area networks (WANs).



Local area networks (wireless and wired)

- Span a small geographic area.
- Usually confined to a building, a group of buildings, or a vehicle, for example a train or a streetcar.
- Data travels between network devices via network cables (LANs) or wireless signals (WLANs).
- The most common types are Ethernet and Wi-Fi.



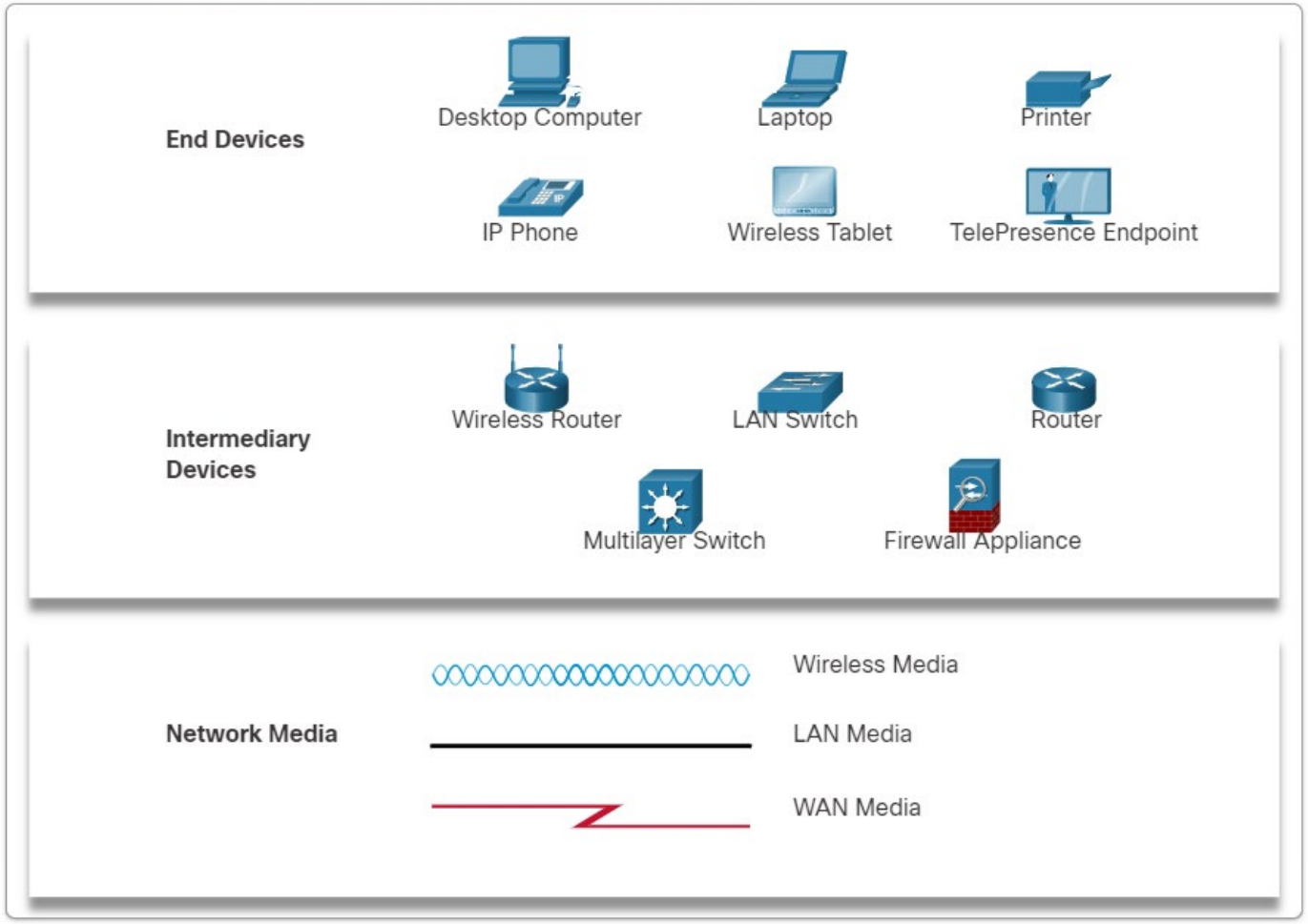
Wide area networks (WAN)

- A WAN is a computer network that covers a large area (any network whose communication links across metropolitan, regional, or national boundaries).
- A network that uses routers, modems, and public communication links.
- The world's largest WAN is the Internet.
- WANs are used to connect LANs and other types of networks together, so that users and computers in one location can communicate with users and computers in other locations.

2. Network Components

Network Infrastructure

- The network infrastructure contains three categories of hardware components.
  - End Devices
  - Intermediary Devices
  - Network Media
- The following icons are used in network topologies to identify a networking component.



End Devices

- The network devices that people are most familiar with are called end devices, or hosts.
  - Computers (workstations, laptops, file servers, web servers)
  - Network printers
  - Telephones and teleconferencing equipment
  - Security cameras
  - Mobile devices (such as smart phones, tablets, PDAs, and wireless debit/credit card readers and barcode scanners)

- An end device (or host) is either the source or destination of a message transmitted over the network.
- To uniquely identify hosts, addresses are used.
- When a host initiates communication, it uses the address of the destination host to specify where the message should be sent.

3. Communication Protocols

Communication Protocols

- Communication in our daily lives takes many forms and occurs in many environments.
- We have different expectations depending on whether we chat via the internet or participate in a job interview.
- Each situation has its corresponding expected behaviors and styles.
- Before beginning to communicate with each other, we establish rules or agreements to govern the conversation that includes the following:
  - What method of communication should we use?
  - What language should we use?
  - Do we need to confirm that our messages are received?



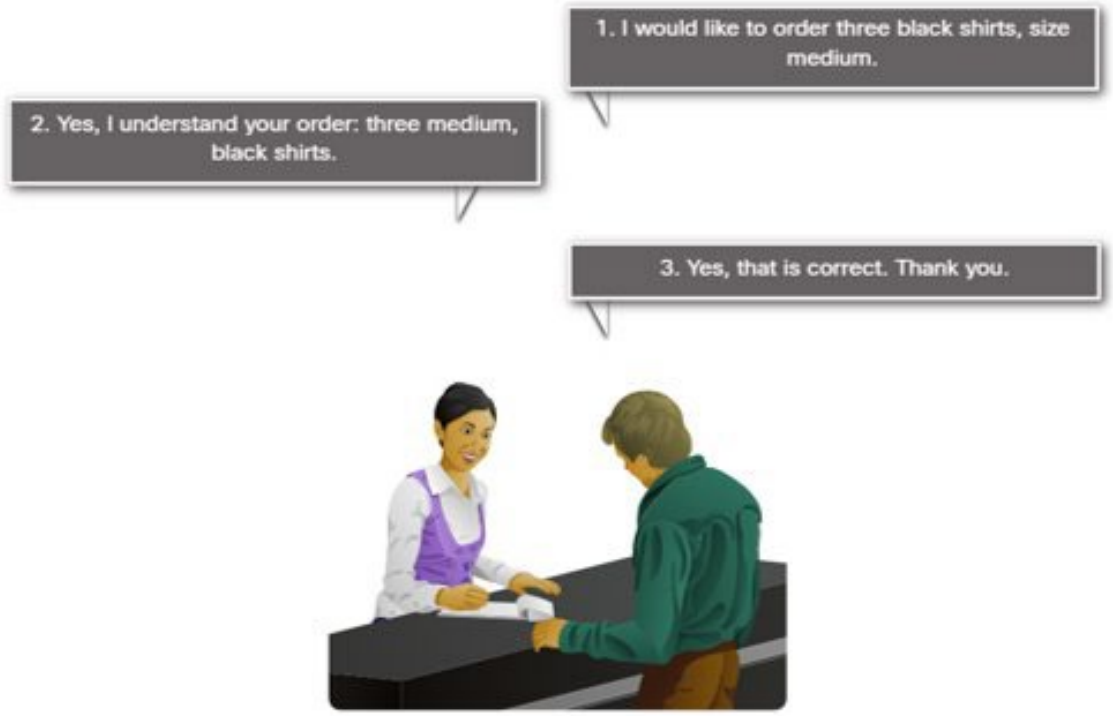
Communication Protocols (Cont.)

Language



Before communication can begin, we may have to reach an agreement on the language used.

Confirmation



Communication is successful when the intended message has been received and confirmed.

Communication Protocols (Cont.)

- These rules, or protocols, must be followed for the message to be successfully delivered and understood.

- Among the protocols that govern successful human communication are these:
  - An identified sender and receiver
  - Agreed upon a method of communicating (face-to-face, telephone, letter, photograph)
  - Common language and grammar
  - Speed and timing of delivery
  - Confirmation or acknowledgment requirements

- The techniques used in network communications share these fundamentals with human conversations.

- Think about the commonly accepted protocols for sending text messages to your friends.

Why Protocols Matter

- Just like humans, computers use rules, or protocols, to communicate.
- Protocols are required for computers to communicate correctly across the network.
- In both a wired and wireless environment, a local network is defined as an area where all hosts must "speak the same language," which, in computer terms, means they must "share a common protocol."
- If everyone in the same room spoke a different language, they would not be able to communicate.
- Likewise, if devices in a local network did not use the same protocols, they would not be able to communicate.
- Networking protocols define many aspects of communication over the local network.

Why Protocols Matter (Cont.)

Protocol Characteristic	Description
Message format	A sent message uses a specific format or structure. It depends on the message type, and the channel used to deliver the message.
Message size	The rules that govern the size of the pieces communicated across the network are stringent. They are different, depending on the channel used. A long message sent from one host to another over a network is necessary to break the message into smaller pieces, ensuring that the message is delivered reliably.
Timing	Many network communication functions are dependent on it. Timing determines the speed at which the bits are transmitted across the network. It also affects when an individual host sends data and the total amount of data sent in any transmission.
Encoding	Messages sent across the network are converted into bits by the sending host. Each bit is encoded into a pattern of sounds, light waves, or electrical impulses depending on the network media over which the bits are transmitted. The destination host receives and decodes the signals to interpret the message.
Encapsulation	Each message transmitted on a network includes a header that contains addressing information that identifies the source and destination hosts; otherwise, it cannot be delivered. Encapsulation is adding this information to the data that make up the message. In addition to addressing, there may be other information in the header that ensures the message is delivered to the correct application on the destination host.
Message pattern	Some messages require an acknowledgment before the following message can be sent. This request/response pattern is a common aspect of many networking protocols. However, other types of messages are streamed across the network without concern about whether they reach their destination.

4. Communication Standards

The Internet and Standards

- With the increasing number of new devices and technologies coming online, how is it possible to manage all the changes and still reliably deliver services such as email?
- The answer is internet standards.
- A standard is a set of rules determining how something must be done.
- Networking and internet standards ensure that all devices connecting to the network implement the same set of rules or protocols in the same manner.
- Different devices send information to each other over the internet using standards.

Network Standards Organizations

- A comprehensive discussion, problem-solving, and testing cycle results in an internet standard.
- These different standards are developed, published, and maintained by various organizations, as shown in the figure.
- A proposed standard's development and approval process are recorded in a numbered RFC document, which tracks the evolution of the standard.
- RFCs for internet standards are published and managed by the IETF.



5. Network Communication Models

The TCP/IP Model

- Layered models help us visualize how the various protocols work together to enable network communications.
- Layered models depict protocol operations occurring within each layer and the interaction with the layers above and below it.
- The layered model has many benefits:
  - It assists protocol design because protocols that operate at a specific layer have defined information that they act upon and a limited interface to the layers above and below.
  - It fosters competition because products from different vendors can work together.
  - Enables technology changes to occur at one level without affecting the other levels.
  - Provides a common language to describe networking functions and capabilities.
- The first layered model for internetwork communications was created in the early 1970s and referred to as the internet model.
- It defines four categories of functions that must occur for communications to be successful.

The TCP/IP Model (Cont.)

- The suite of TCP/IP protocols used for internet communications follows the structure of this model, as shown in the table.

- Because of this, the internet model refers to the TCP/IP model.

TCP/IP Model Layer	Description
Application	Represents data to the user, plus encoding and dialog control.
Transport	Supports communication between various devices across diverse networks.
Internet	Determines the best path through the network.
Network Access	Controls the hardware devices and media that make up the network.

The OSI Reference Model

- There are two basic types of models that we use to describe the functions that must occur for network communications to be successful:
    - **Protocol model** - It closely matches the structure of a particular protocol suite. A protocol suite includes a set of related protocols that typically provide all the functionality required for people to communicate with the data network. The TCP/IP model is a protocol model because it describes the functions at each layer of protocols within the TCP/IP suite.
    - **Reference model** – It describes the functions that must be completed at a particular layer but needs to specify exactly how a function should be accomplished. This model does not intend to provide sufficient detail to define precisely how each protocol should work at each layer. The primary purpose of a reference model is to aid in a clearer understanding of the functions and processes necessary for network communications.
- The OSI project at the ISO created the most widely known internetwork reference model.
- It is used for data network design, operation specifications, and troubleshooting.
- This model refers to the OSI model.

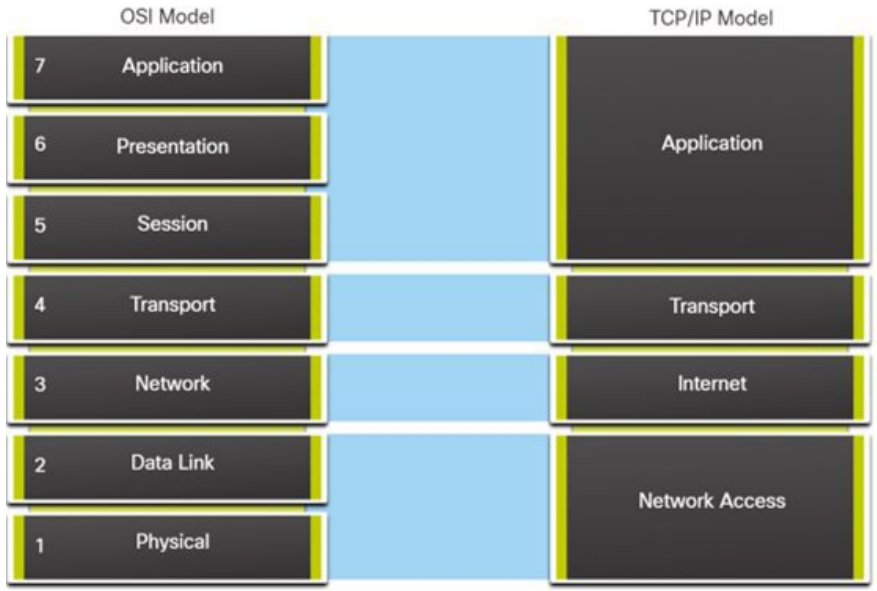
The OSI Reference Model (Cont.)

OSI Model Layer	Description
7 - Application	The application layer contains protocols used for process-to-process communications.
6 - Presentation	The presentation layer provides a typical representation of the data transferred between application layer services.
5 - Session	The session layer provides services to the presentation layer to organize its dialogue and manage data exchange.
4 - Transport	The transport layer defines services to segment, transfer, and reassemble the data for individual communications between the end devices.
3 - Network	The network layer provides services to exchange individual data over the network between identified end devices.
2 - Data Link	The data link layer protocols describe methods for exchanging data frames between devices over shared media.
1 - Physical	The physical layer protocols describe the mechanical, electrical, functional, and procedural means to activate, maintain, and de-activate physical connections for a bit transmission to and from a network device.

OSI Model and TCP/IP Model Comparison

- Because TCP/IP is the protocol suite used for internet communications, why do we need to learn the OSI model as well?
- The TCP/IP model visualizes the interactions of the various protocols that make up the TCP/IP protocol suite.
- It does not describe general functions that are necessary for all networking communications.
- It describes the networking functions specific to those protocols in the TCP/IP protocol suite.
- The protocols that make up the TCP/IP protocol suite describes in terms of the OSI reference model.

- The functions that occur at the internet layer in the TCP/IP model are contained in the network layer of the OSI Model, as shown in the figure.
- The transport layer functionality is the same between both models.



OSI Model and TCP/IP Model Comparison (Cont.)

- The key similarities are in the transport and network layers; however, the two models differ in how they relate to the layers above and below each layer:
  - OSI Layer 3, the network layer, maps directly to the TCP/IP internet layer. This layer describes protocols that address and route messages through the internetwork.
  - OSI Layer 4, the transport layer, maps directly to the TCP/IP transport layer. This layer describes general services and functions that deliver ordered and reliable data between source and destination hosts.
  - The TCP/IP application layer includes several protocols that provide specific functionality to various end-user applications. Application software developers and vendors use the OSI model Layers 5, 6, and 7 as references to produce applications that operate on networks.
  - The TCP/IP and OSI models commonly refer to protocols at various layers because the OSI model separates the data link layer from the physical layer, which commonly refers to these lower layers.

6. Common Application Protocol

Common Application Protocols

- Telnet
- File Transfer Protocol (FTP)
- Trivial File Transfer Protocol (TFTP)
- Simple Mail Transfer Protocol (SMTP)
- Post Office Protocol (POP3)
- Internet Message Access Protocol (IMAP)
- Dynamic Host Configuration Protocol (DHCP)

Common Application Protocols (Contd.)

- Hypertext Transfer Protocol (HTTP)
- Secure Sockets Layer (SSL)/ Transport Layer Security (TLS)
- Network File System (NFS)
- Simple Network Management Protocol (SNMP)

Implications of Multilayer Protocols

- Encapsulation
  - [Ethernet [IP [TCP [HTTP]
  - [Ethernet [IP [TCP [SSL [HTTP]
  - [Ethernet [IPSec [IP [TCP [SSL [HTTP]]
  - [Ethernet [IP [TCP [HTTP [FTP]]
  - [Ethernet [IP [ICMP [TCP [HTTP]]
- Double encapsulation, VLAN hopping
- Encryption, flexibility, resiliency
- Covert channels, filter bypass, segmentation violations

7. IPv4 Address

The IPv4 Address

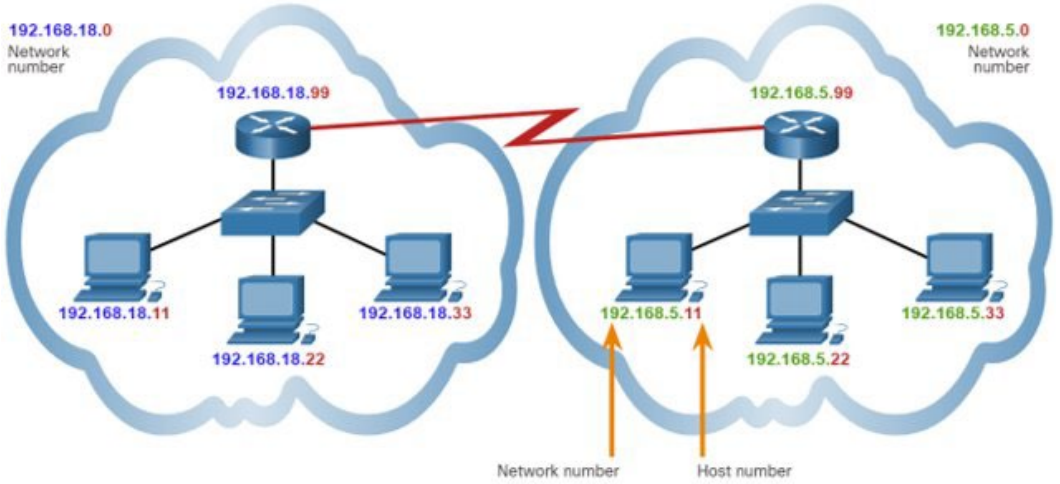
- A host needs an IPv4 address to participate on the internet and almost all LANs today.
- The IPv4 address is a logical network address that identifies a particular host.
- It must be configured appropriately and uniquely within the LAN for local communication.
- It must also be appropriately configured and unique worldwide for remote communication.
- This is how a host can communicate with other devices on the internet.
- An IPv4 address is assigned to the network interface connection for a host that is usually a NIC installed in the device.
- Examples of end-user devices with network interfaces include workstations, servers, network printers, and IP phones.
- Some servers can have more than one NIC, each of which has its IPv4 address.
- Router interfaces that connect to an IP network will also have an IPv4 address.
- Every packet sent across the internet has a source and destination IPv4 address.
- Networking devices require this information to ensure it gets to the destination and returns any replies to the source.

Octets and Dotted-Decimal Notation

- IPv4 addresses are 32 bits in length.
- Here is an IPv4 address in binary: 11010001101001011100100000000001
- Imagine having to configure devices with a series of 32 bits, which groups the 32 bits into four 8-bit bytes called octets like this: 11010001.10100101.11001000.00000001
- That's better but still difficult to read, so we convert each octet into its decimal value, separated by a decimal point or period.
- The above binary IPv4 becomes this dotted-decimal representation: 209.165.200.1

Networks and Hosts

- Two parts make up the IPv4 address, the network (blue) and the host (red).
- An IPv4 address requires both parts.
- Both networks have the subnet mask 255.255.255.0 used to identify the network which connects the host.
- The figure shows a host with an IPv4 address 192.168.5.11 with a subnet mask of 255.255.255.0.
- The first three octets (192.168.5) identify the network portion of the address, and the last octet (11) identifies the host.
- This is known as hierarchical addressing because the network portion indicates the network which locates each unique host address.
- Routers only need to know how to reach each network rather than needing to see the location of each host.



Networks and Hosts (Cont.)

- With IPv4 addressing, multiple logical networks can exist on one physical network if the network portion of the logical network host addresses is different.
- For example, three hosts on a single, physical local network have the same IPv4 address (192.168.18), and three other hosts have different network portions of their IPv4 addresses (192.168.5).
- The hosts with the same network number in their IPv4 addresses will be able to communicate with each other but will not be able to communicate with the other hosts without the use of routing.
- This example shows one physical network and two logical IPv4 networks.
- Another example of a hierarchical network is the telephone system.
- With a telephone number, the country code, area code, and exchange represent the network address, and the remaining digits represent a local phone number.

Public and Private IPv4 Addresses

- Just as there are different ways to transmit an IPv4 packet, there are also different types of IPv4 addresses. Some IPv4 addresses cannot be used to go out to the internet, and others are allocated explicitly for routing to the internet. Some are used to verify a connection, and others are self-assigned.

- Internet service provider (ISP) routers globally route public IPv4 addresses. However, not all available IPv4 addresses can be used on the internet. There are blocks of addresses called private addresses that most organizations use to assign IPv4 addresses to internal hosts.

- In the mid-1990s, with the introduction of the World Wide Web (WWW), private IPv4 addresses were introduced because of the depletion of IPv4 address space. Private IPv4 addresses are not unique and can be used internally within any network.

- The long-term solution to IPv4 address depletion was IPv6.

Public and Private IPv4 Addresses (Cont.)

- Private addresses are defined in RFC 1918 and are sometimes referred to as RFC 1918 address space.

Network Address and Prefix	RFC 1918 Private Address Range
10.0.0.0/8	10.0.0.0 - 10.255.255.255
172.16.0.0/12	172.16.0.0 - 172.31.255.255
192.168.0.0/16	192.168.0.0 - 192.168.255.255

Special Use IPv4 Addresses

- Specific addresses, such as network and broadcast addresses, cannot be assigned to hosts.
- Special addresses can also be set to hosts, but with restrictions on how those hosts can interact within the network.

Link-Local addresses

- Link-local addresses (169.254.0.0 /16 or 169.254.0.1 to 169.254.255.254) are more commonly known as the Automatic Private IP Addressing (APIPA) addresses or self-assigned addresses.
- A Windows client uses them to self-configure if the client cannot obtain an IP address through other methods.
- Link-local addresses can be used in a peer-to-peer connection but are not commonly used for this purpose.

Special Use IPv4 Addresses (Cont.)

Loopback addresses

- Loopback addresses (127.0.0.0 /8 or 127.0.0.1 to 127.255.255.254) commonly identify as only 127.0.0.1.
- These are special addresses used by a host to direct traffic to itself.
- For example, test connections frequently use the ping command to other hosts. (Ping will be covered later in this course.)
- But you can also use the ping command to test the IP configuration on your device, as shown in the figure.

Pinging the Loopback Interface

```
C:\Users\NetAcad> ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Ping statistics for 127.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\NetAcad> ping 127.1.1.1
Pinging 127.1.1.1 with 32 bytes of data:
Reply from 127.1.1.1: bytes=32 time<1ms TTL=128
Reply from 127.1.1.1: bytes=32 time<1ms TTL=128
Reply from 127.1.1.1: bytes=32 time<1ms TTL=128
Reply from 127.1.1.1: bytes=32 time<1ms TTL=128
Ping statistics for 127.1.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\NetAcad>
```

Legacy Classful Addressing

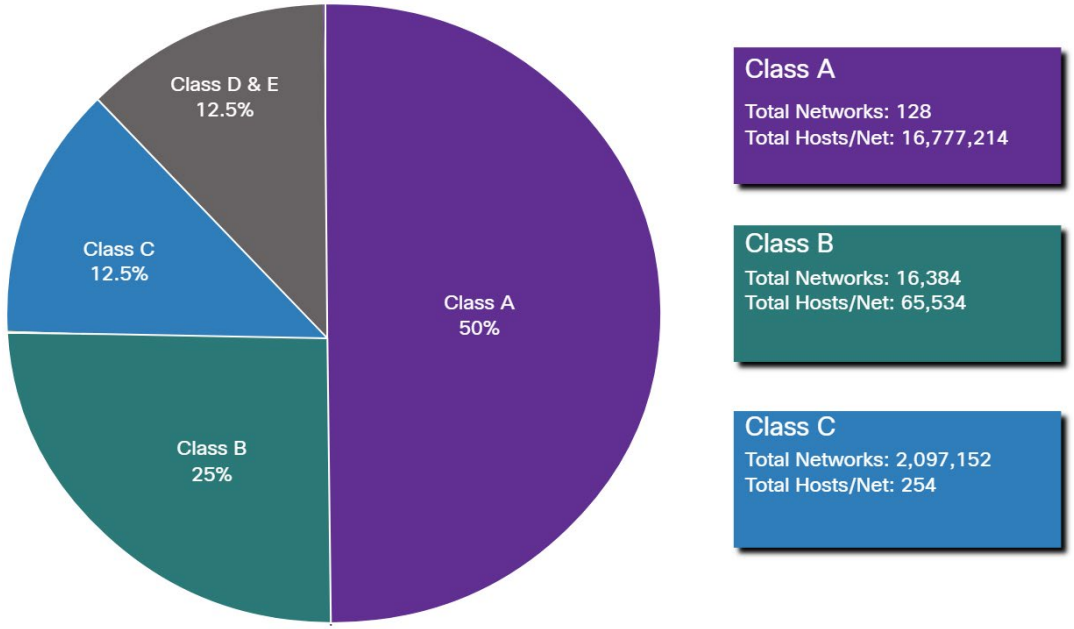
In 1981, IPv4 addresses were assigned using classful addressing defined in RFC 790 (https://tools.ietf.org/html/rfc790), Assigned Numbers. Customers were allocated a network address based on one of three classes, A, B, or C. The RFC divided the unicast ranges into specific classes as follows:

- **Class A (0.0.0.0/8 to 127.0.0.0/8)** - Designed to support vast networks with more than 16 million host addresses. Class A used a fixed /8 prefix with the first octet to indicate the network address and the remaining three octets for host addresses (more than 16 million host addresses per network).
- **Class B (128.0.0.0 /16 - 191.255.0.0 /16)** - Designed to support the needs of moderate to large-size networks with up to approximately 65,000 host addresses. Class B used a fixed /16 prefix with the two high-order octets to indicate the network address and the remaining two for host addresses (more than 65,000 host addresses per network).
- **Class C (192.0.0.0 /24 - 223.255.255.0 /24)** - Designed to support small networks with a maximum of 254 hosts. Class C used a fixed /24 prefix with the first three octets to indicate the network and the remaining octet for the host addresses (only 254 host addresses per network).

Legacy Classful Addressing (Cont.)

Note: There is also a Class D multicast block consisting of 224.0.0.0 to 239.0.0.0 and a Class E experimental address block consisting of 240.0.0.0 - 255.0.0.0.

- At the time, with a limited number of computers using the internet, classful addressing was an effective means to allocate addresses. As shown in the figure, Class A and B networks have many host addresses, and Class C has very few. Class A networks accounted for 50% of the IPv4 networks. This caused most of the available IPv4 addresses to go unused.



Legacy Classful Addressing (Cont.)

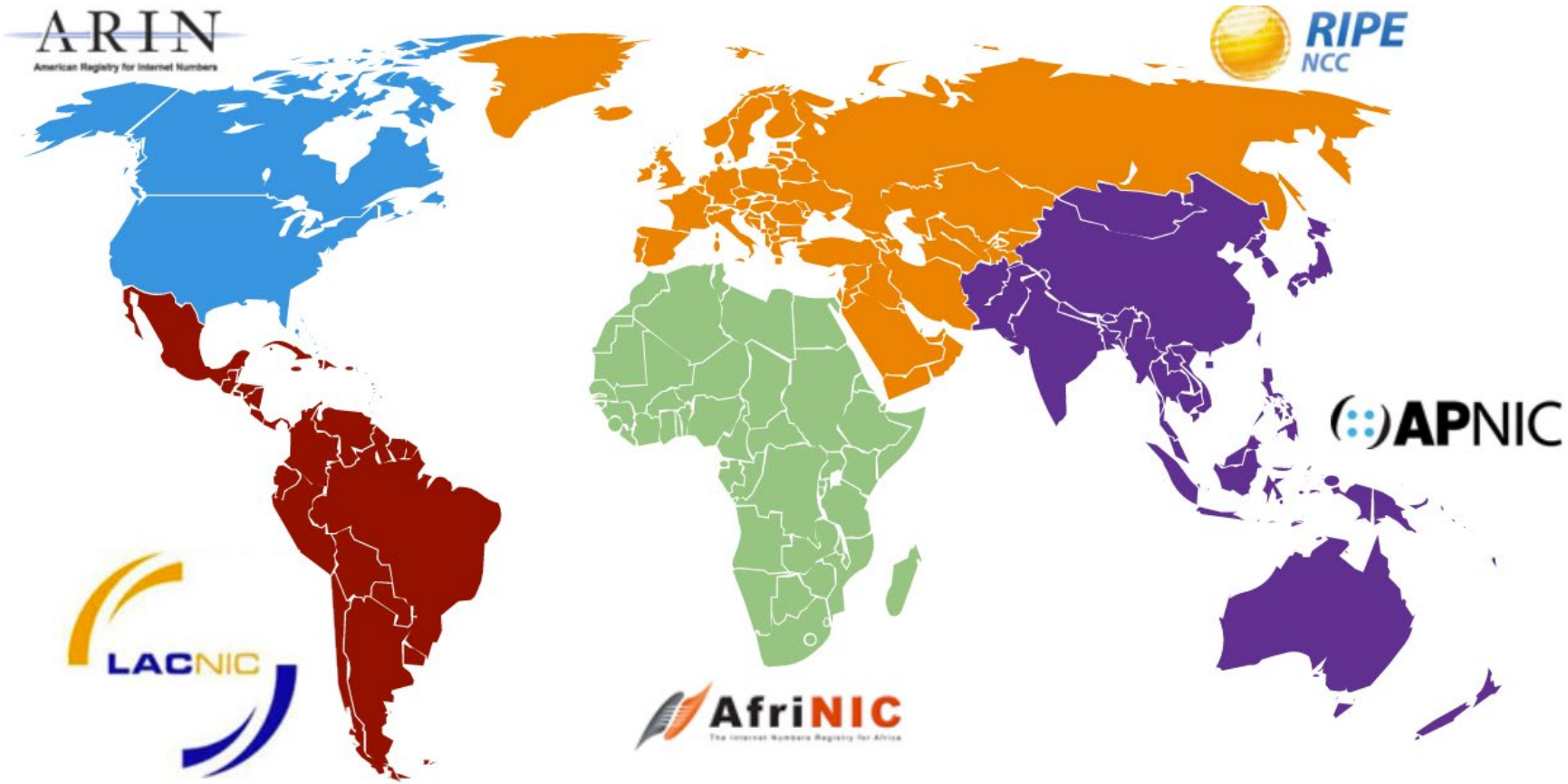
- In the mid-1990s, with the introduction of the World Wide Web (WWW), classful addressing was deprecated to more efficiently allocate the limited IPv4 address space.
- Classful address allocation was replaced with classless addressing, which is used today.
- Classless addressing ignores the rules of classes (A, B, C).
- Public IPv4 network addresses (network addresses and subnet masks) are allocated based on the number of addresses that can be justified.

Assignment of IP Addresses

- The internet globally routes public IPv4 addresses. These addresses must be unique.
- The Internet Assigned Numbers Authority manages IPv4 and IPv6 addresses (IANA).
- The IANA manages and allocates blocks of IP addresses to the Regional Internet Registries (RIRs).
- RIRs allocate IP addresses to ISPs that provide IPv4 address blocks to organizations and smaller ISPs.
- Organizations can also get their addresses directly from an RIR (subject to the policies of that RIR).

Assignment of IP Addresses (Cont.)

Regional Internet Registries



Assignment of IP Addresses (Cont.)

- **AfriNIC** (African Network Information Centre) - Africa Region
- **APNIC** (Asia Pacific Network Information Centre) - Asia/Pacific Region
- **ARIN** (American Registry for Internet Numbers) - North America Region
- **LACNIC** (Regional Latin-American and Caribbean IP Address Registry) - Latin America and some Caribbean Islands
- **RIPE NCC** (Réseaux IP Européens Network Coordination Centre) - Europe, the Middle East, and Central Asia

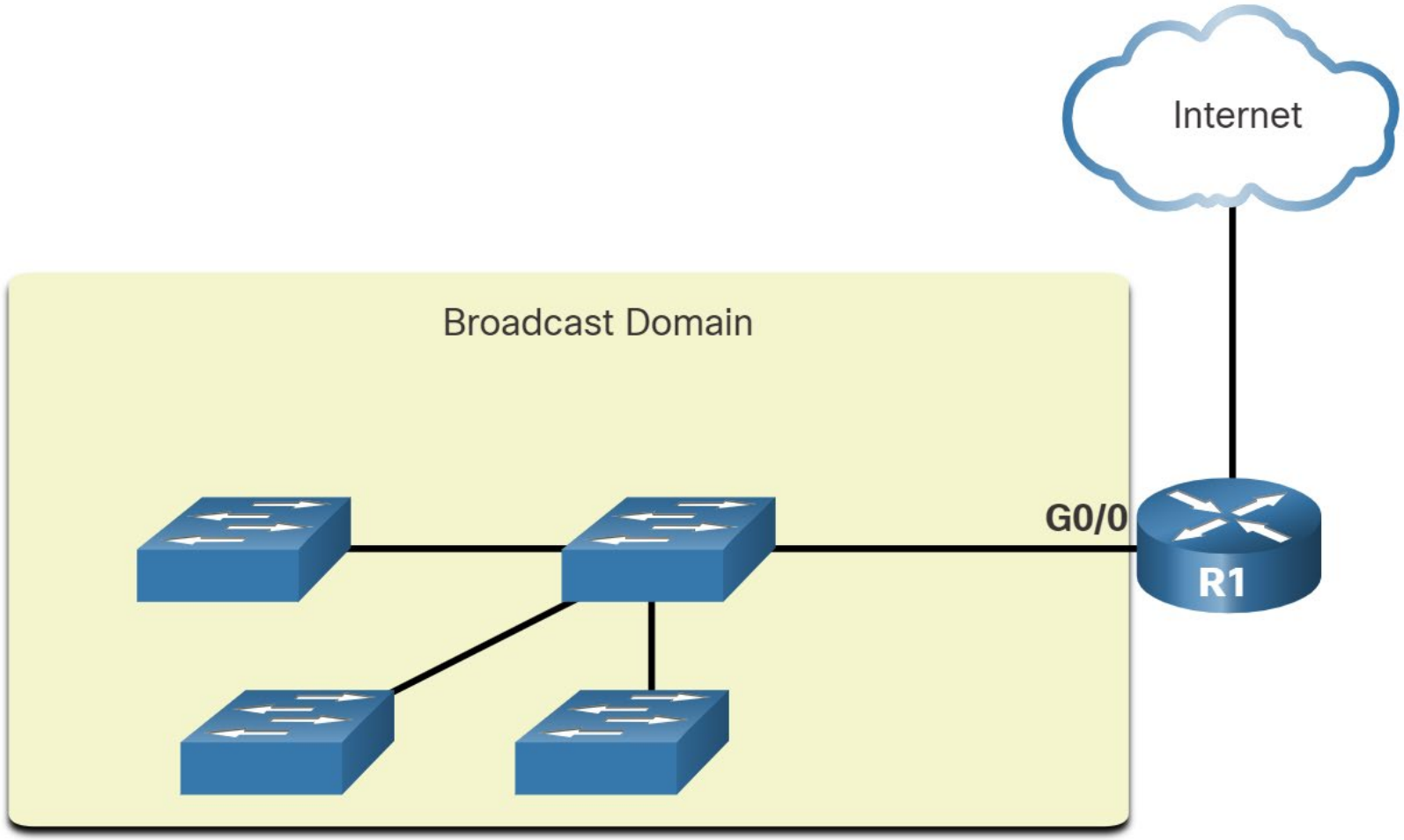
8. Network Segmentation

Broadcast Domains and Segmentation

- In an Ethernet LAN, devices use broadcasts and the Address Resolution Protocol (ARP) to locate other devices.
- ARP sends Layer 2 broadcasts to a known IPv4 address on the local network to discover the MAC address.
- Devices on Ethernet LANs also find other devices using services.
- A host typically acquires its IPv4 address configuration using the Dynamic Host Configuration Protocol (DHCP), which sends broadcasts on the local network to locate a DHCP server.
- Switches propagate broadcasts out all interfaces except the interface on which it was received.
- For example, if a switch in the figure were to receive a broadcast, it would forward it to the other switches and users connected to the network.

Broadcast Domains and Segmentation (Cont.)

Routers Segment Broadcast Domains

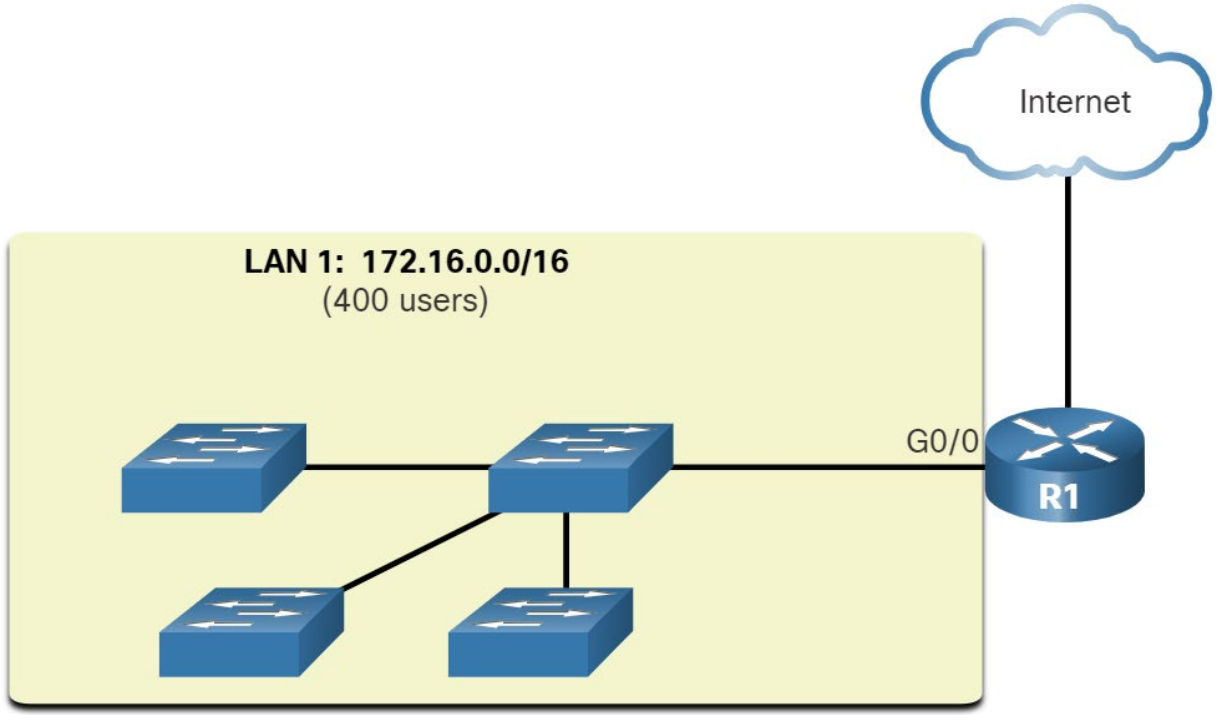


Broadcast Domains and Segmentation (Cont.)

- Routers do not propagate broadcasts.
- When a router receives a broadcast, it does not forward it to other interfaces.
- For instance, when R1 receives a broadcast on its Gigabit Ethernet 0/0 interface, it does not forward out another interface.
- Therefore, each router interface connects to a broadcast domain and only propagates within that specific broadcast domain.

Problems with Large Broadcast Domains

- A large broadcast domain is a network that connects many hosts.
- A problem with a large broadcast domain is that these hosts can generate excessive broadcasts and negatively affect the network.
- In the figure, LAN 1 connects 400 users, which could generate excessive broadcast traffic.
- This results in slow network operations due to the significant amount of traffic it can cause and slow device operations because a device must accept and process each broadcast packet.



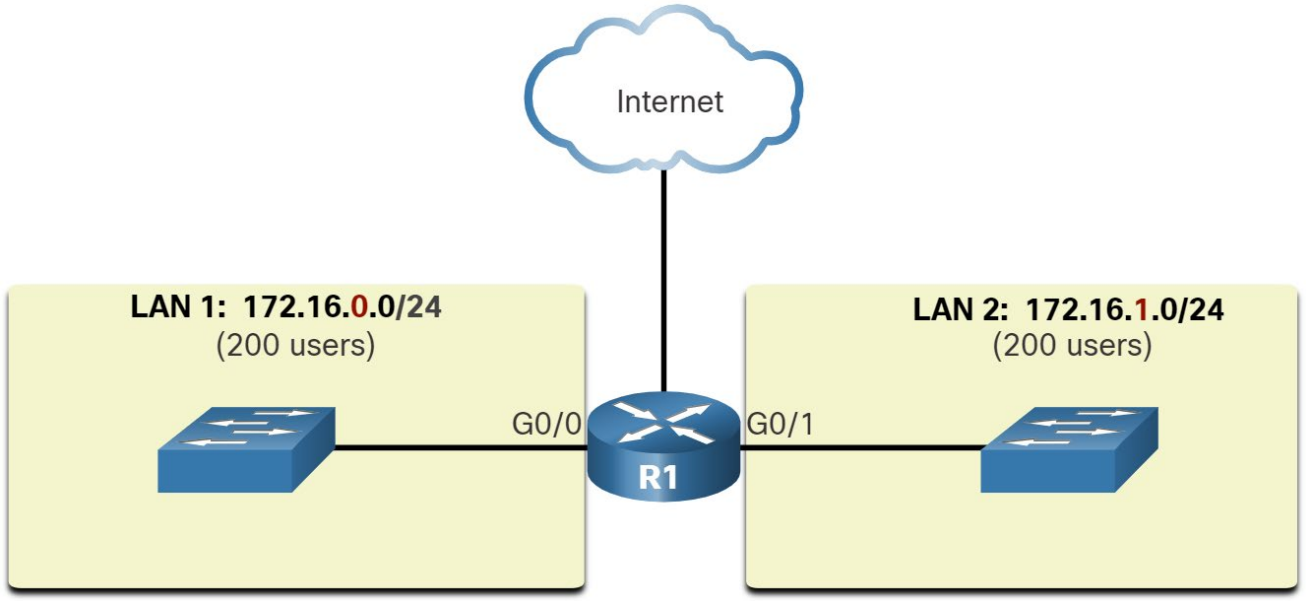
Problems with Large Broadcast Domains (Cont.)

- The solution is to reduce the network size to create smaller broadcast domains in a subnetting process. These smaller network spaces are called subnets.

- In the figure, the 400 users in LAN 1 with network address 172.16.0.0 /16 have been divided into two subnets of 200 users each: 172.16.0.0 /24 and 172.16.1.0 /24.

- Broadcasts only propagate within the smaller broadcast domains.

- Therefore, a broadcast in LAN 1 would not propagate to LAN 2.



Problems with Large Broadcast Domains (Cont.)

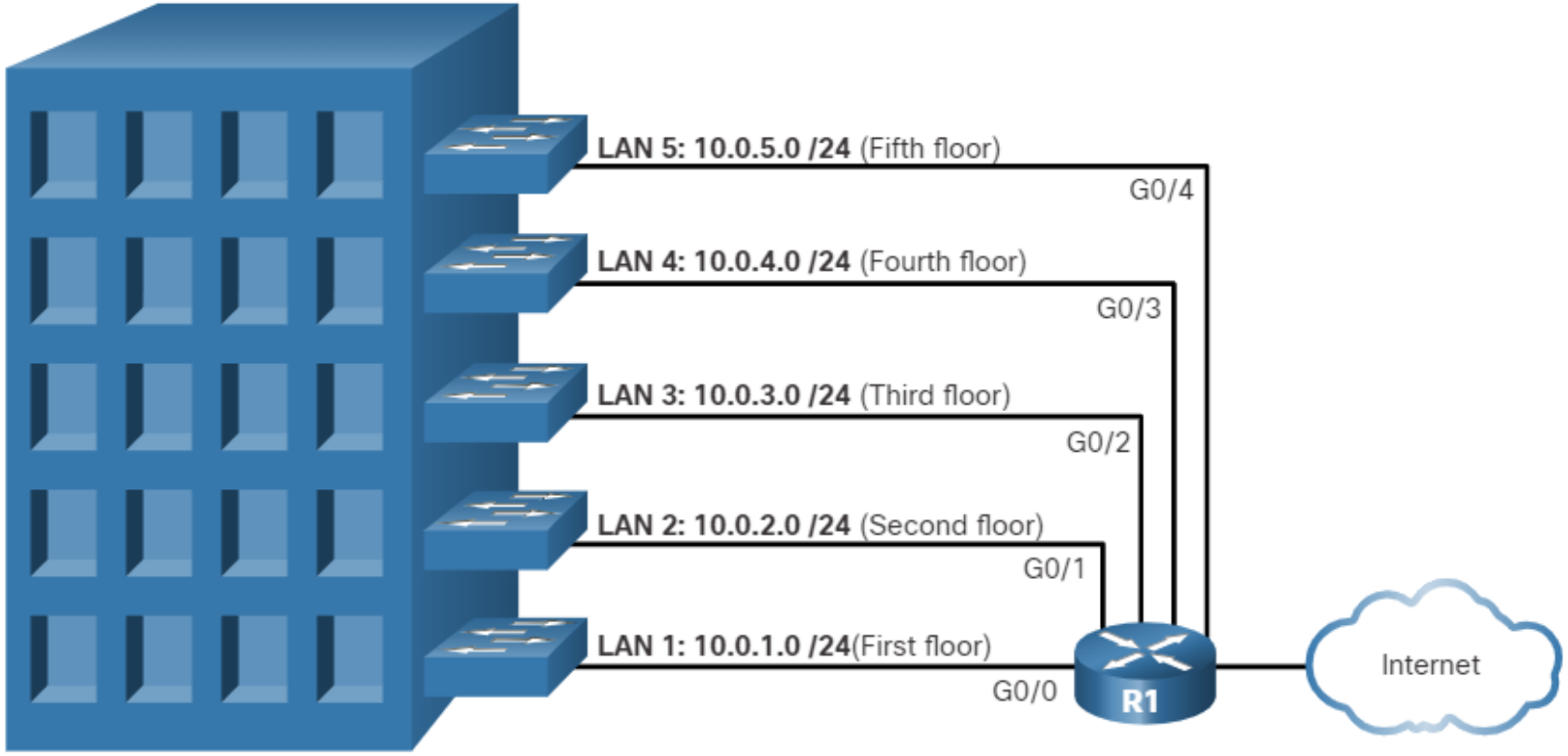
- Notice how the prefix length has changed from a single /16 network to two /24 networks.
- This is the basis of subnetting: using host bits to create additional subnets.
- The use of the terms subnet and network are interchangeable.
- Most networks are a subnet of some larger address block.

Reasons for Segmenting Networks

- Subnetting reduces overall network traffic and improves network performance.
- It also enables an administrator to implement security policies, such as which subnets are allowed or not allowed to communicate together.
- Another reason is that it reduces the number of devices affected by abnormal broadcast traffic due to misconfigurations, hardware/software problems, or malicious intent.
- There are various ways of using subnets to help manage network devices.

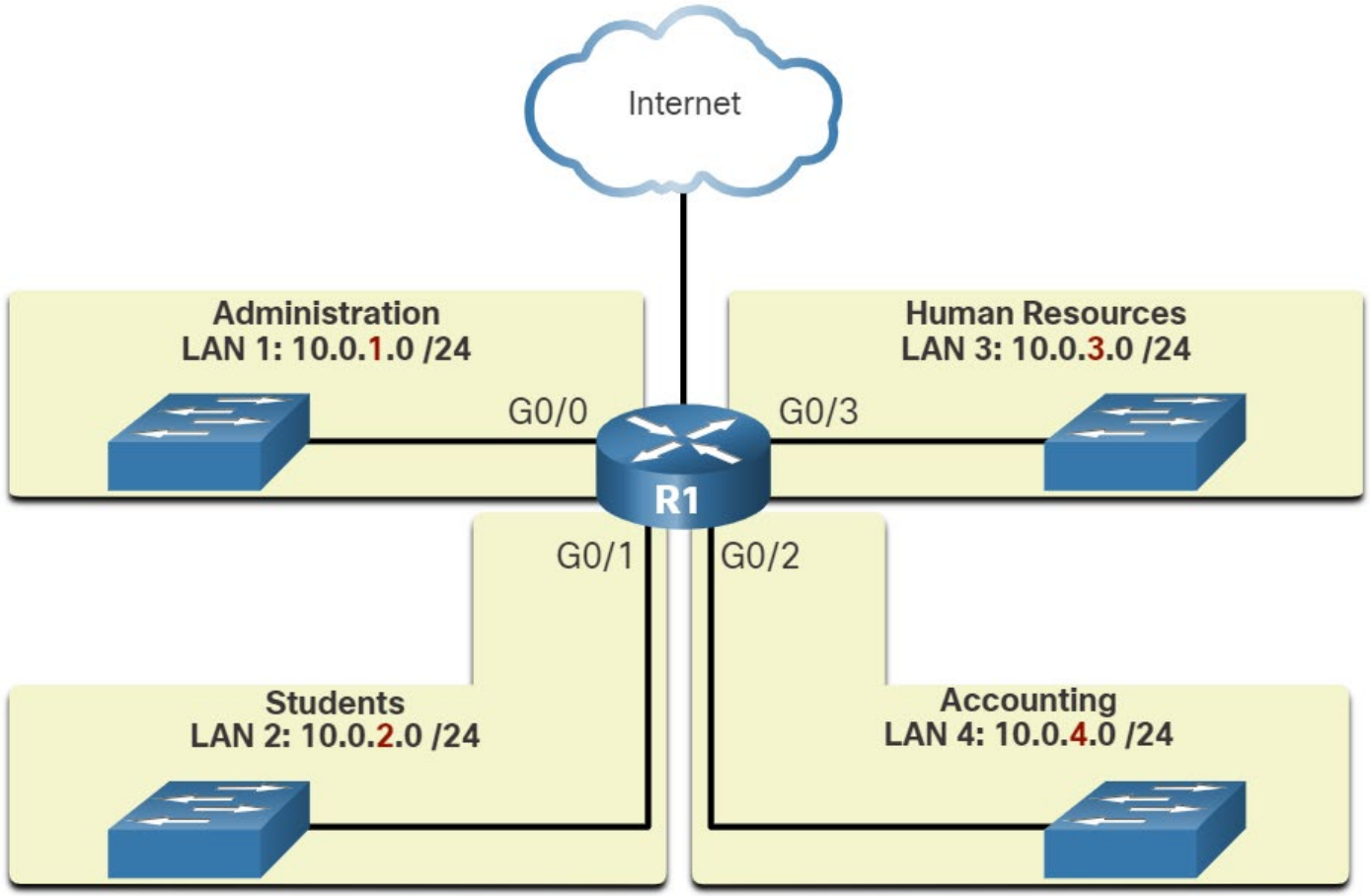
Reasons for Segmenting Networks (Cont.)

Subnetting by Location



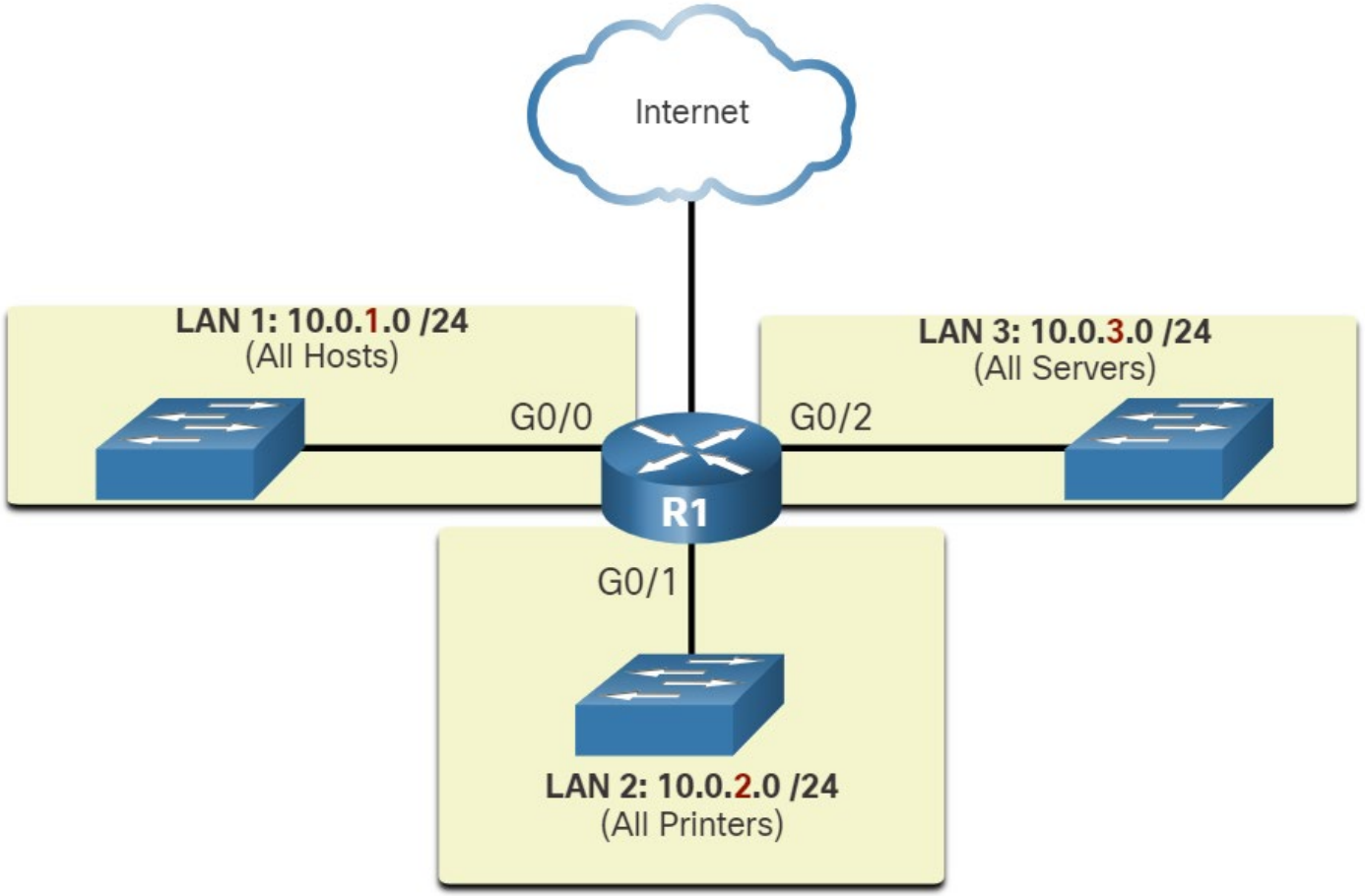
Reasons for Segmenting Networks (Cont.)

Subnetting by Group or Function



Reasons for Segmenting Networks (Cont.)

Subnetting by Device Type



Reasons for Segmenting Networks (Cont.)

- Network administrators can create subnets using any other division that makes sense for the network.
- Notice in each figure the subnets use longer prefix lengths to identify networks.
- Understanding how to subnet networks is a fundamental skill that all network administrators must develop.
- This process creates various methods to understand it better.
- Although a little overwhelming initially, pay close attention to the detail, and subnetting will become easier with practice.

9. IPv6 Addressing

Hexadecimal Number System

- Before diving into IPv6 addressing, it's important to know that IPv6 addresses are represented using hexadecimal numbers.

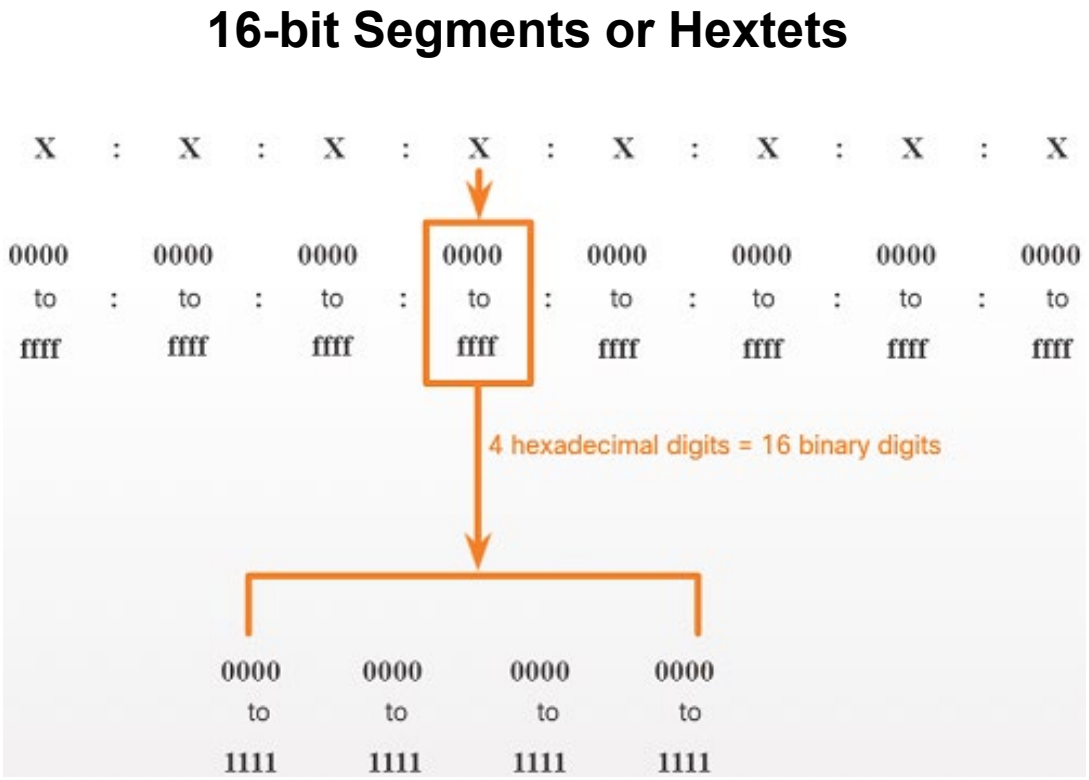
- This base sixteen number system uses the digits 0 to 9 and the letters A to F:

0 1 2 3 4 5 6 7 8 9 A B C D E F

- In IPv6 addresses, these 16 digits are represented as hextets (discussed next) allowing us to represent these massive addresses in a much more readable format.

IPv6 Addressing Formats

- The first step to learning about IPv6 in networks is to understand the way an IPv6 address is written and formatted.
- IPv6 addresses are much larger than IPv4 addresses, which is why we are unlikely to run out of them.
- IPv6 addresses are 128 bits in length and written as a string of hexadecimal values.
- Every four bits is represented by a single hexadecimal digit; for a total of 32 hexadecimal values, as shown in the figure.
- IPv6 addresses are not case-sensitive and can be written in either lowercase or uppercase.



IPv6 Addressing Formats (Cont.)

- The previous figure also shows that the preferred format for writing an IPv6 address is x:x:x:x:x:x:x:x, with each “x” consisting of four hexadecimal values.
- The term octet refers to the eight bits of an IPv4 address.
- In IPv6, a hextet is the unofficial term used to refer to a segment of 16 bits, or four hexadecimal values.
- Each “x” is a single hextet which is 16 bits or four hexadecimal digits.
- Preferred format means that you write IPv6 address using all 32 hexadecimal digits.
- It does not necessarily mean that it is the ideal method for representing the IPv6 address.
- In this module, you will see two rules that help to reduce the number of digits needed to represent an IPv6 address.

Examples of IPv6 Addresses in the Preferred Format

```
2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0000: 0200
2001 : 0db8 : 0000 : 00a3 : abcd : 0000 : 0000: 1234
2001 : 0db8 : 000a : 0001 : c012 : 9aff : fe9a: 19ac
2001 : 0db8 : aaaa : 0001 : 0000 : 0000 : 0000: 0000
fe80 : 0000 : 0000 : 0000 : 0123 : 4567 : 89ab: cdef
fe80 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000: 0001
fe80 : 0000 : 0000 : 0000 : c012 : 9aff : fe9a: 19ac
fe80 : 0000 : 0000 : 0000 : 0123 : 4567 : 89ab: cdef
0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000: 0001
0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000: 0001
```

Rule 1 – Omit Leading Zeros

- The first rule to help reduce the notation of IPv6 addresses is to omit any leading 0s (zeros) in any hextet.

- Here are four examples of ways to omit leading zeros:
  - 01ab can be represented as 1ab
  - 09f0 can be represented as 9f0
  - 0a00 can be represented as a00
  - 00ab can be represented as ab

- This rule only applies to leading 0s, NOT to trailing 0s, otherwise the address would be ambiguous.

- For example, the hextet “abc” could be either “0abc” or “abc0”, but these do not represent the same value.

Type	Format
Preferred	2001 : **0**db8 : **000**0 : 1111 : **000**0 : **000**0 : **000**0 : **0**200
No leading 0s	2001 : db8 : 0 : 1111 : 0 : 0 : 0 : 200
Preferred	2001 : **0**db8 : **000**0 : **00**a3 : ab00 : **0**ab0 : **00**ab : 1234
No leading 0s	2001 : db8 : 0 : a3 : ab00 : ab0 : ab : 1234
Preferred	2001 : **0**db8 : **000**a : **000**1 : c012 : 90ff : fe90 : **000**1
No leading 0s	2001 : db8 : a : 1 : c012 : 90ff : fe90 : 1
Preferred	2001 : **0**db8 : aaaa : **000**1 : **000**0 : **000**0 : **000**0 : **000**0
No leading 0s	2001 : db8 : aaaa : 1 : 0 : 0 : 0 : 0
Preferred	fe80 : **000**0 : **000**0 : **000**0 : **0**123 : 4567 : 89ab : cdef
No leading 0s	fe80 : 0 : 0 : 0 : 123 : 4567 : 89ab : cdef
Preferred	fe80 : **000**0 : **000**0 : **000**0 : **000**0 : **000**0 : **000**0 : **000**1
No leading 0s	fe80 : 0 : 0 : 0 : 0 : 0 : 0 : 1
Preferred	**000**0 : **000**0 : **000**0 : **000**0 : **000**0 : **000**0 : **000**0 : **000**1
No leading 0s	0 : 0 : 0 : 0 : 0 : 0 : 0 : 1
Preferred	**000**0 : **000**0 : **000**0 : **000**0 : **000**0 : **000**0 : **000**0 : **000**0
No leading 0s	0 : 0 : 0 : 0 : 0 : 0 : 0 : 0

Rule 2- Double Colon

- The second rule to help reduce the notation of IPv6 addresses is that a double colon (::) can replace any single, contiguous string of one or more 16-bit hextets consisting of all zeros.
- For example, 2001:db8:cafe:1:0:0:0:1 (leading 0s omitted) could be represented as 2001:db8:cafe:1::1.
- The double colon (::) can only be used once within an address, otherwise there would be more than one possible resulting address.
- When used with the omitting leading 0s technique, the notation of IPv6 address can often be greatly reduced.
- This is commonly known as the compressed format.

Rule 2- Double Colon (Cont.)

- Here is an example of the incorrect use of the double colon: 2001:db8::abcd::1234 because it is used twice.

- Here are the possible expansions of this incorrect compressed format address:
  - 2001:db8::abcd:0000:0000:1234
  - 2001:db8::abcd:0000:0000:0000:1234
  - 2001:db8:0000:abcd::1234
  - 2001:db8:0000:0000:abcd::1234

- If an address has more than one contiguous string of all-0 hextets, best practice is to use the double colon (::) on the longest string.

- If the strings are equal, the first string should use the double colon (::).

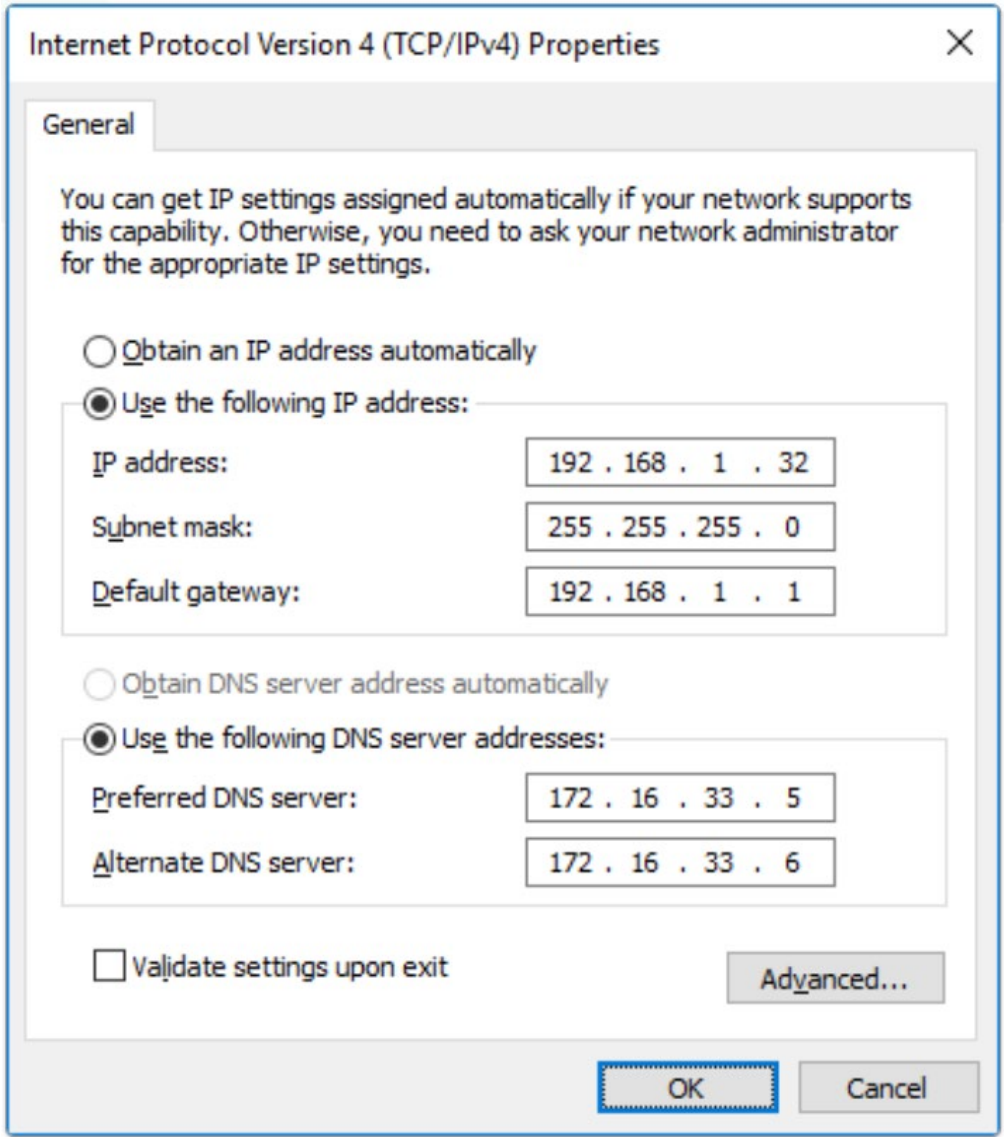
Type	Format
Preferred	2001 : 0db8 : 0000 : 1111 : 0000 : 0000 : 0000 : 0200
Compressed/spaces	2001 : db8 : 0 : 1111 : : 200
Compressed	2001:db8:0:1111::200
Preferred	2001 : 0db8 : 0000 : 0000 : ab00 : 0000 : 0000 : 0000
Compressed/spaces	2001 : db8 : 0 : 0 : ab00 : :
Compressed	2001:db8:0:0:ab00::
Preferred	2001 : 0db8 : aaaa : 0001 : 0000 : 0000 : 0000 : 0000
Compressed/spaces	2001 : db8 : aaaa : 1 : :
Compressed	2001:db8:aaaa:1::
Preferred	fe80 : 0000 : 0000 : 0000 : 0123 : 4567 : 89ab : cdef
Compressed/spaces	fe80 : : 123 : 4567 : 89ab : cdef
Compressed	fe80::123:4567:89ab:cdef
Preferred	fe80 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0001
Compressed/spaces	fe80 : : 1
Compressed	fe80::1
Preferred	0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0001
Compressed/spaces	: : 1
Compressed	::1
Preferred	0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000 : 0000
Compressed/spaces	: :
Compressed	::

10. Static and Dynamic Addressing

Static IPv4 Address Assignment

- IPv4 addresses can be assigned either statically or dynamically.

- Statically assigning an IP configuration on a device requires the manual entry of:
  - **IP address** - Identifies the host on the network.
  - **Subnet mask** – Identifies the network address.
  - **Default gateway** – Identifies the gateway of last resort.

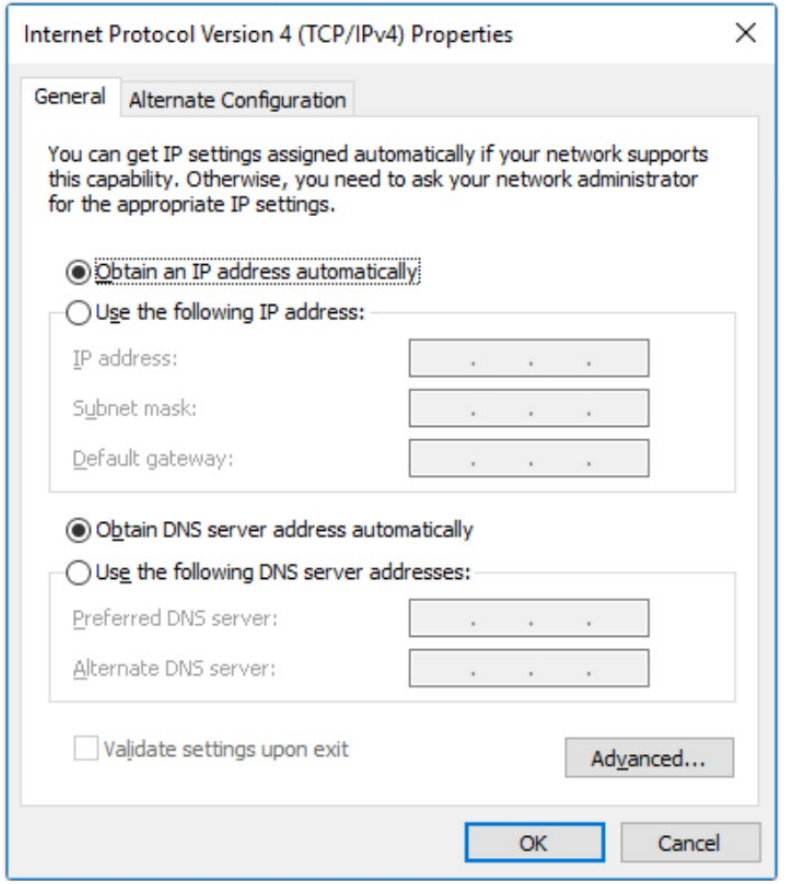


Static IPv4 Address Assignment (Cont.)

- Static addresses are useful for printers, servers, and other networking devices that need to be accessible to clients on the network.
- If hosts normally access a server at a particular IPv4 address, it would not be good if that address changed.
- Static assignment of addressing information can provide increased control of network resources, but it can be time consuming to enter the information on each host.
- When IPv4 addresses are entered statically, the host only performs basic error checks on the IPv4 address. Therefore, errors are more likely to occur.
- When using static IPv4 addressing, it is important to maintain an accurate list of which IPv4 addresses are assigned to which devices.
- These are permanent addresses and are not normally reused.

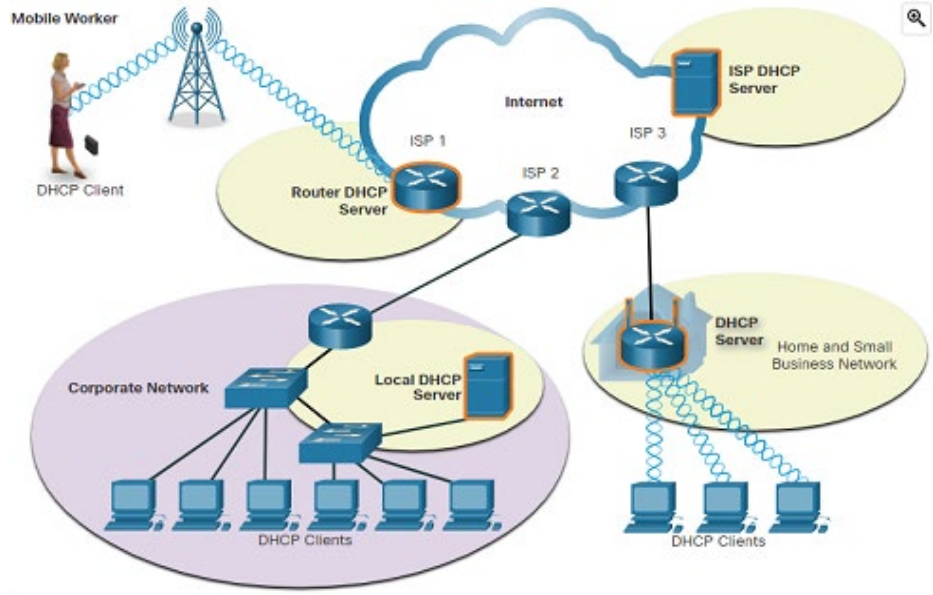
Dynamic IPv4 Address Assignment

- Dynamic Host Configuration Protocol (DHCP) dynamically assigns IPv4 address configurations to end devices.
- DHCP is commonly used to automatically assign IP configurations to connecting hosts on large networks because it reduces the burden on network support staff and virtually eliminates entry errors.
- IP configurations are “leased” for a limited amount of time.
- When a host disconnects from a network, the IP address is returned to the DHCP pool for reuse.



Dynamic Host Configuration Protocol

- As the figure shows, various types of devices can be DHCP servers.
- The DHCP server in most medium-to-large networks is usually a local, dedicated PC-based server.
- With home networks, the location of the DHCP server is on the local router that connects the home network to the ISP.
- Many networks use both DHCP and static addressing.
- The use of the DHCP is for general-purpose hosts, such as end-user devices.

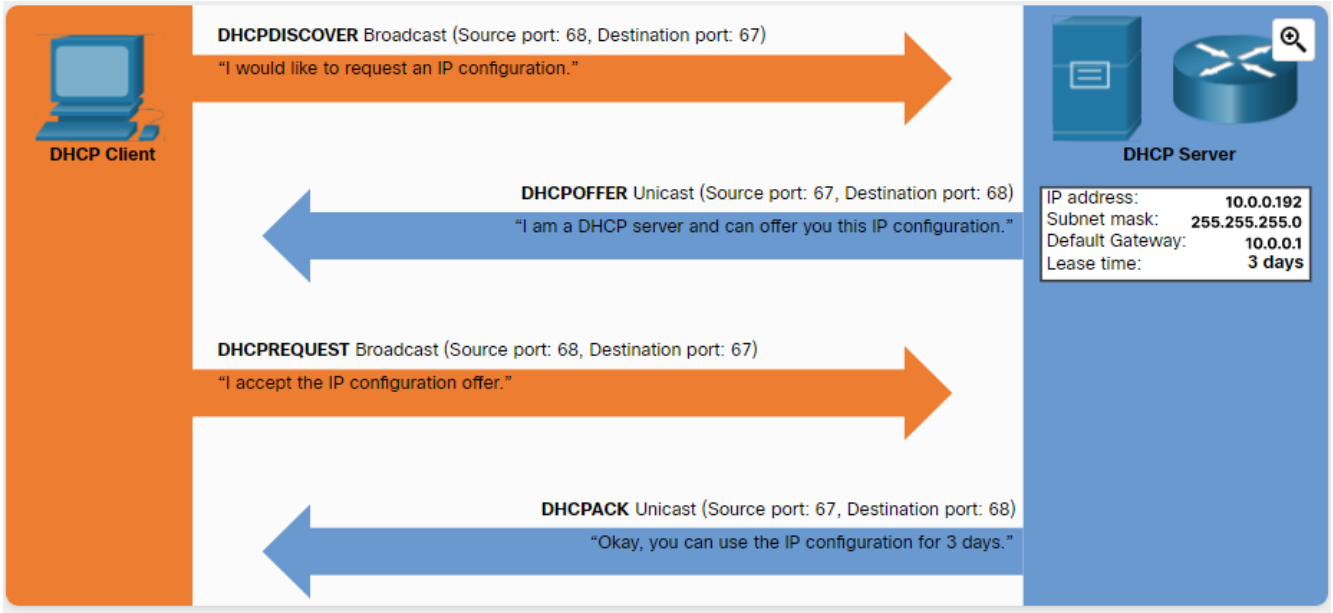


- The static addressing is for network devices, such as gateway routers, switches, servers, and printers.
- DHCPv6 provides similar services for IPv6 clients, but it does not provide a default gateway address.

Network Services

DHCP Server (Cont.)

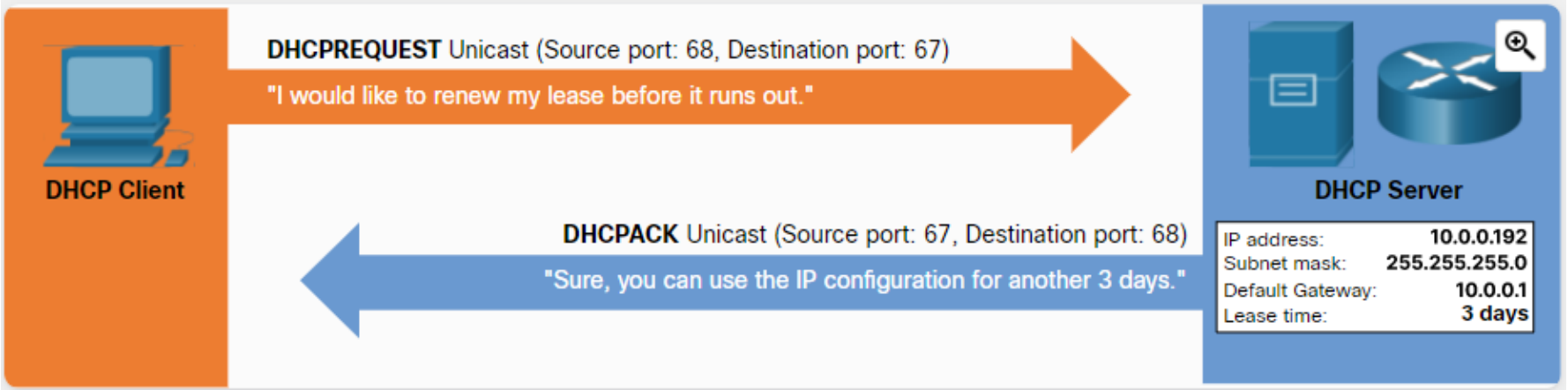
- DHCP works in a client/server mode, where DHCP clients request available IP configurations from a DHCP server.
- A DHCP server is configured with a scope (i.e., a pool or a range) of addresses that it can lease to requesting DHCP clients.
- As shown in the figure, when the DHCP client boots (or otherwise wants to join a network), it initiates the following four-step process to obtain a lease.



Network Services

DHCP Server (Cont.)

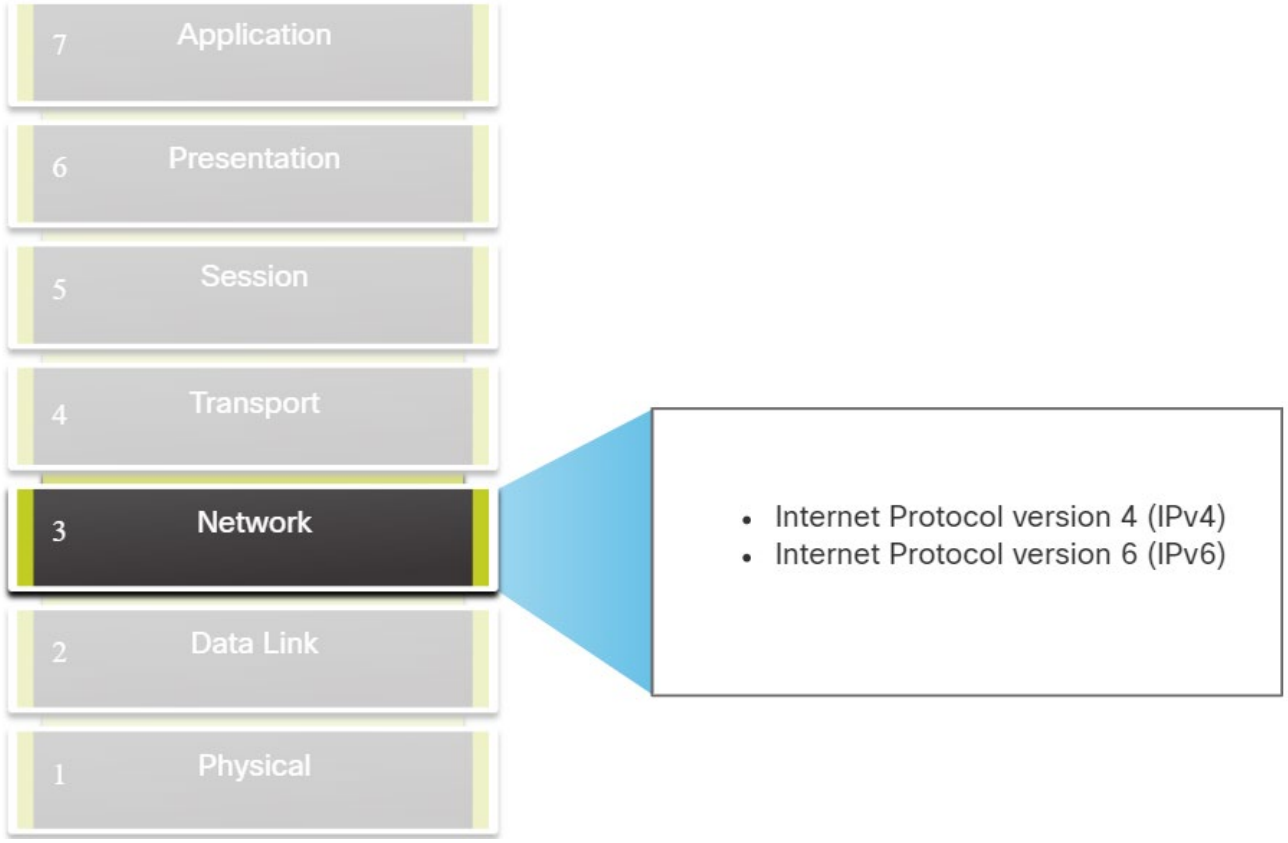
- The client must contact the DHCP server periodically to extend the lease.
- This lease mechanism ensures that moved or power-off clients do not keep addresses that they no longer need.
- When a lease expires, the DHCP server returns the address to the pool where it can be reallocated as necessary.



11. Network Layer or Internet Layer

The Network Layer

- The network layer, or OSI Layer 3, provides services to allow end devices to exchange data across networks.
- IP version 4 (IPv4) and version 6 (IPv6) are the principal network layer communication protocols.
- Other network layer protocols include routing protocols such as Open Shortest Path First (OSPF) and messaging protocols such as Internet Control Message Protocol (ICMP).



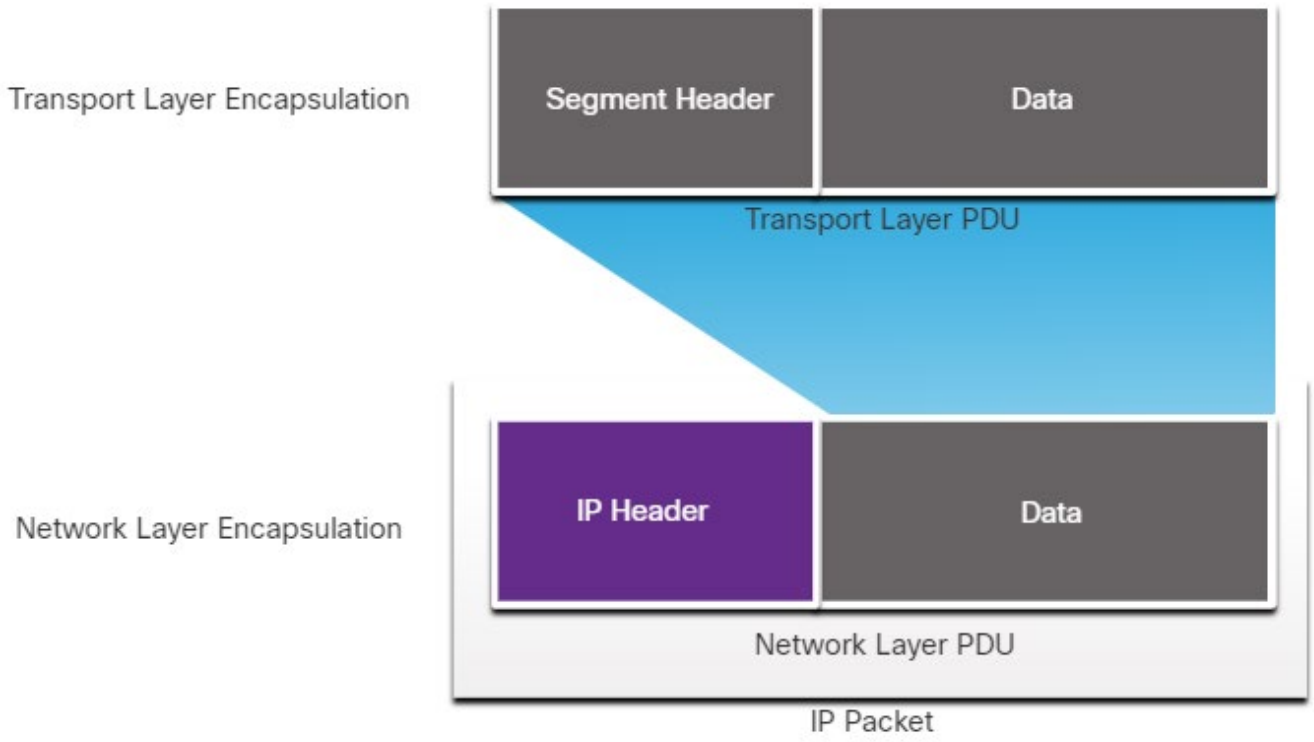
The Network Layer (Cont.)

Network layer protocols perform four basic operations to accomplish end-to-end communications across network boundaries:

- **Addressing end devices** - A unique IP address for identification on the network must configure end devices.
- **Encapsulation** - The network layer encapsulates the protocol data unit (PDU) from the transport layer into a packet.
- **Routing** - The network layer provides services to direct the packets to a destination host on another network. A router must process the packet to travel to other networks. The role of the router is to select the best path and direct packets toward the destination host in a process known as routing.
- **De-encapsulation** - When the packet arrives at the network layer of the destination host, the packet is de-encapsulated by the network layer, and the appropriate service at the transport layer passes up the resulting Layer 4 PDU. The destination host of the IP packet performs the de-encapsulation process.

IP Encapsulation

- IP encapsulates the transport layer (just above the network layer) segment or other data by adding an IP header.
- Delivering the packet to the destination host uses the IP header.



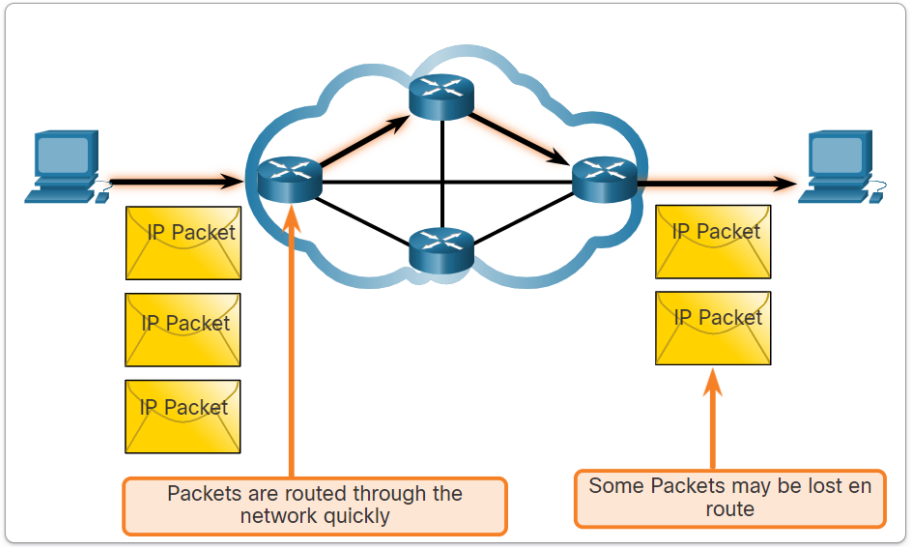
IP Encapsulation (Cont.)

- Encapsulating data layer by layer enables the services at the different layers to develop and scale without affecting the other layers.
- IPv4 or IPv6, or any new protocol that develops in the future, can package the transport layer segments readily.
- The IP header is examined by Layer 3 devices (i.e., routers and Layer 3 switches) as it travels across a network to its destination.
- The IP addressing information remains the same from when the packet leaves the source host until it arrives at the destination host, except when translated by the device performing NAT for IPv4.
- Routers implement routing protocols to route packets between networks.
- The encapsulated transport layer PDU or other data remains unchanged during the network layer processes.

Characteristics of IP

- IP was designed as a protocol with low overhead.
- It provides only the functions necessary to deliver a packet from a source to a destination over an interconnected system of networks.
- The protocol's design is not to track and manage the flow of packets.
- These functions, if required, are performed by other protocols at other layers, primarily TCP at Layer 4.

- These are the essential characteristics of IP:
  - **Connectionless:** There is no connection with the destination established before sending data packets.
  - **Best Effort:** IP is inherently unreliable because packet delivery is not guaranteed.
  - **Media Independent:** Operation is independent of the medium (i.e., copper, fiber-optic, or wireless) carrying the data.



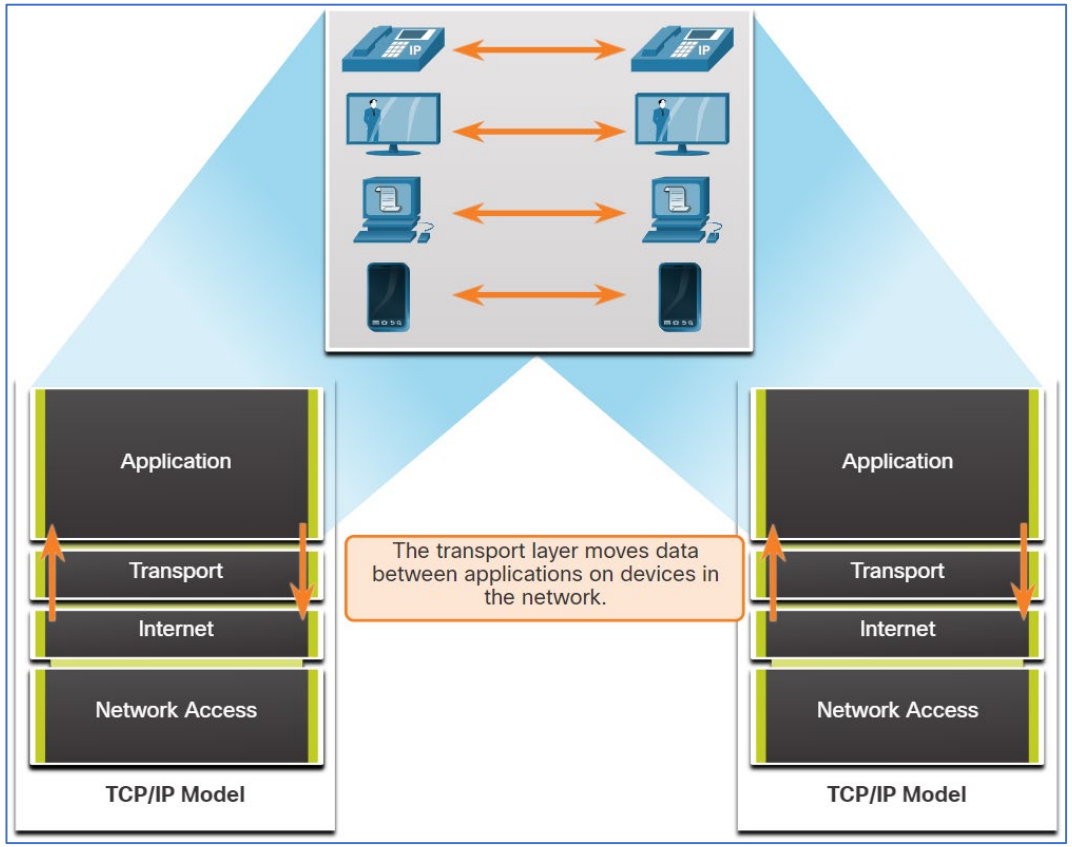
12. Transport Layer

Role of the Transport Layer

- Application layer programs generate data that must be exchanged between source and destination hosts.

- The transport layer is responsible for logical communications between applications running on different hosts.
  - It is the link between the application layer and the lower network transmission layers.

- The transport layer includes two protocols:
  - Transmission Control Protocol (TCP)
  - User Datagram Protocol (UDP)

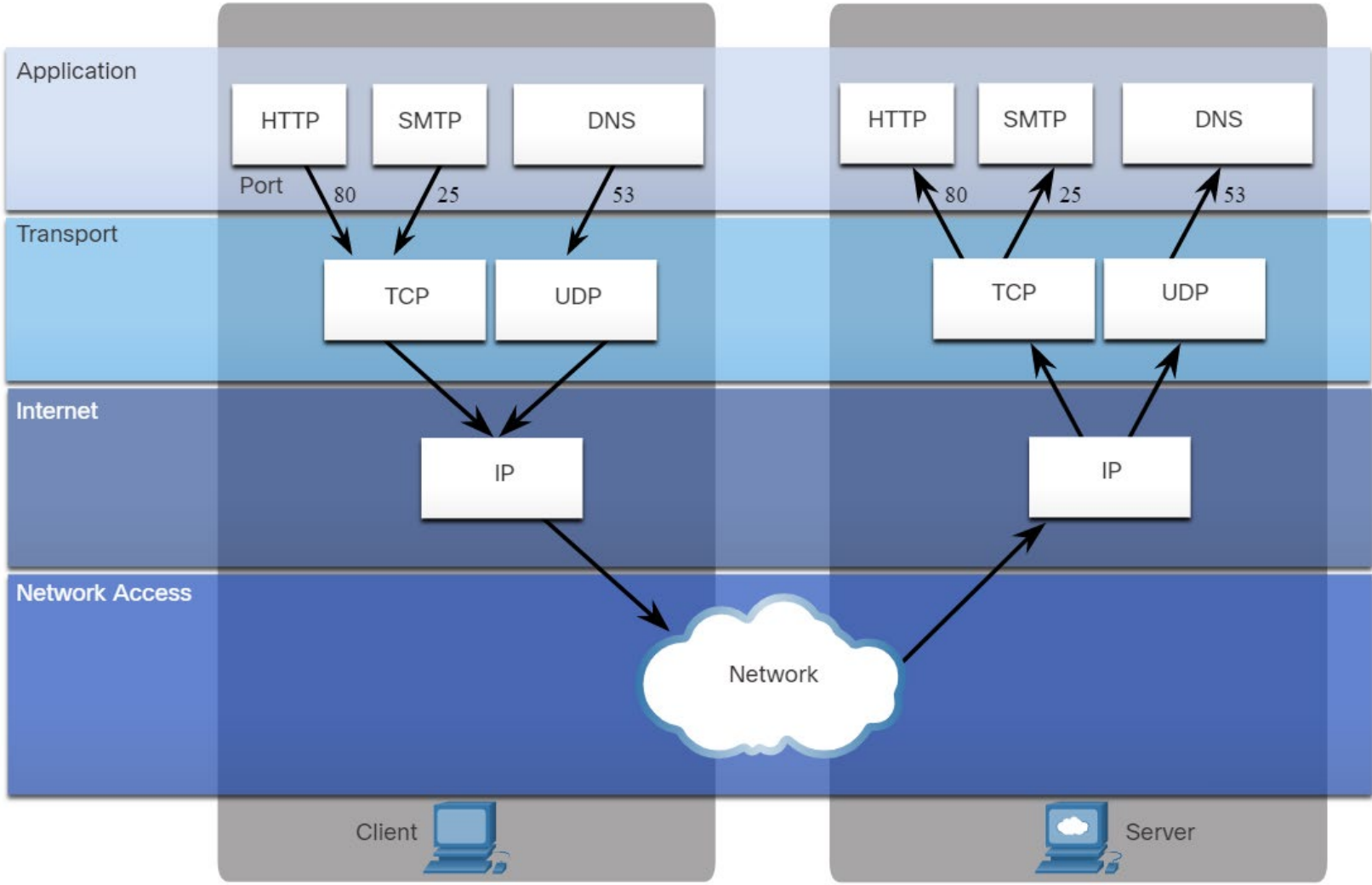


Transport Layer Responsibilities

- The transport layer responsibilities include:
  - **Tracking individual conversations** - Each set of data flowing between a source application and a destination application is known as a conversation and is tracked separately.
  - **Segmenting data and reassembling segments** - The transport layer divides the application data into smaller sized segments (or datagrams) making it easier to manage and transport. The receiving host transport layer reassembles the segments / datagrams.
  - **Adding header information** – The transport layer adds several fields in a header.
  - **Identifying applications** - The transport layer identifies the target application using an identifier called a port number.
  - **Conversation multiplexing** - The transport layer uses segmentation and multiplexing to enable different communication conversations to be interleaved on the same network.

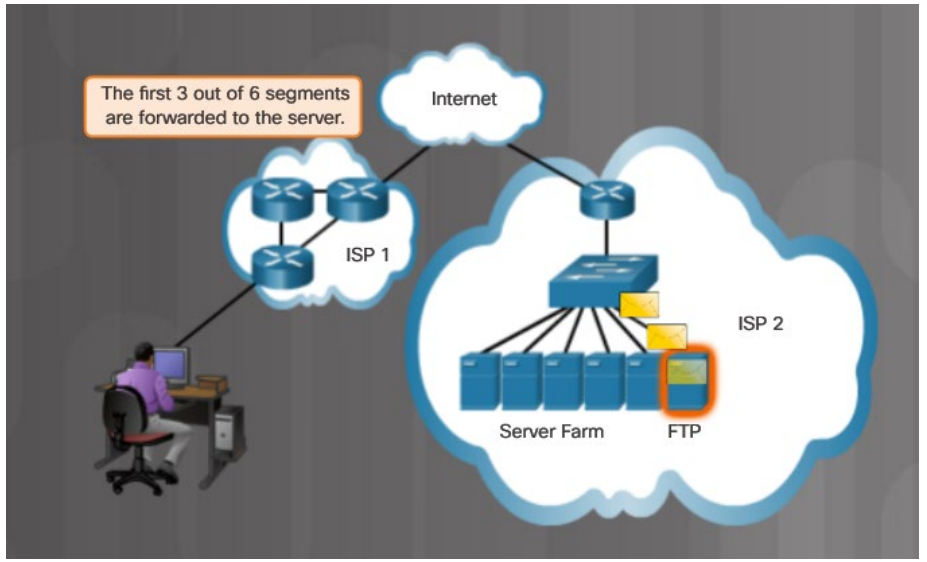
Transport Layer Protocols

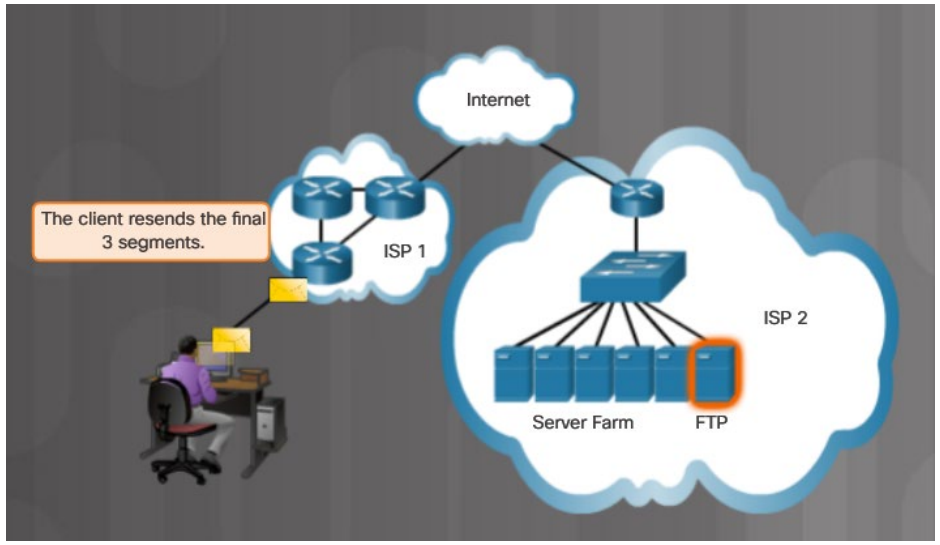
- Transport layer protocols specify how to transfer messages between hosts.
- They are responsible for managing reliability requirements of a conversation.
- Different applications have different transport reliability requirements.
- Therefore, TCP/IP provides the TCP and UDP protocols.



Transport Layer Protocols
TCP

- Three basic operations of reliability
  - Numbering and tracking of data segments
  - Acknowledgment of received data
  - Retransmitting any unacknowledged data after a period of time.

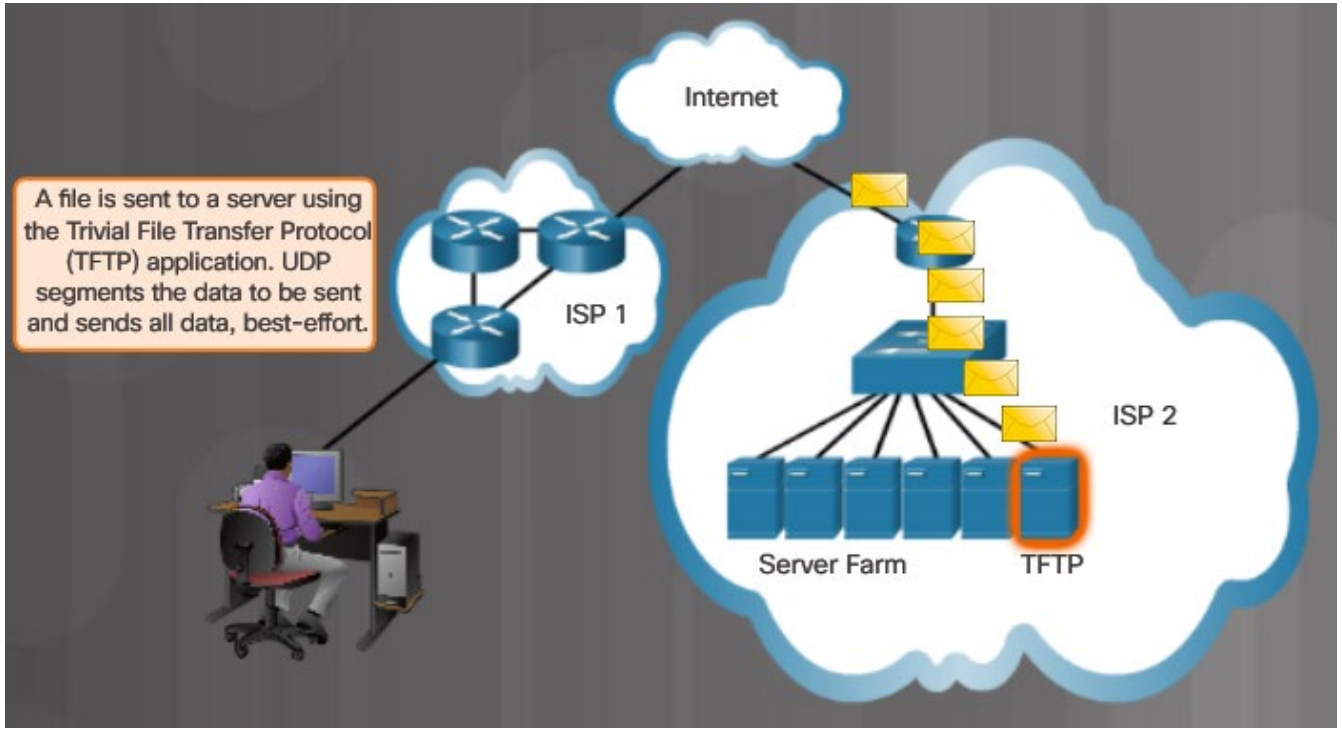




Transport Layer Protocols

UDP

- Very little overhead or data checking
- Best-effort delivery protocol (unreliable)
  - No acknowledgment that the data is received by the destination



TCP and UDP Port Numbers

- When a server receives a message, the server must determine which service is requested by the client. A destination port preconfigures a client to use it that registers on the internet for each service. An example is web browser clients preconfigured to send requests to web servers using port 80, the well-known port for HTTP web services.

- Ports are assigned and managed by an organization known as the Internet Corporation for Assigned Names and Numbers (ICANN). Ports are broken into three categories and range in number from 1 to 65,535:
  - **Well-Known Ports** - Destination ports associated with typical network applications identify as well-known ports. These ports are in the range of 1 to 1023.
  - **Registered Ports** - Either source or destination ports are the use of ports 1024 through 49151. Organizations can use these to register specific applications such as IM applications.
  - **Private Ports** - Often, source ports are the use of ports 49152 through 65535. Any application can use these ports.

TCP and UDP Port Numbers (Cont.)

- Some applications may use both TCP and UDP. For example, DNS uses UDP when clients send requests to a DNS server. However, communication between two DNS servers always uses TCP.

- Search the IANA website for the port registry to view the full port numbers and associated applications list.

TCP and UDP Port Numbers (Cont.)

Port Number	Transport	Application Protocol
20	TCP	File Transfer Protocol (FTP) - Data
21	TCP	FTP - Control
22	TCP	Secure Shell (SSH)
23	TCP	Telnet
25	TCP	Simple Mail Transfer Protocol (SMTP)
53	UDP, TCP	Domain Name System (DNS)
67	UDP	Dynamic Host Configuration Protocol (DHCP) - Server
68	UDP	DHCP - Client
69	UDP	Trivial File Transfer Protocol (TFTP)
80	TCP	Hypertext Transfer Protocol (HTTP)
110	TCP	Post Office Protocol version 3 (POP3)
143	TCP	Internet Message Access Protocol (IMAP)
161	UDP	Simple Network Management Protocol (SNMP)
443	TCP	Hypertext Transfer Protocol Secure (HTTPS)

The netstat Command

- Unexplained TCP connections can pose a significant security threat. They indicate that something or someone connects to the local host. Sometimes it is necessary to know which active TCP connections are open and running on a networked host. Netstat is an important network utility that can be used to verify those connections. As shown below, enter the command netstat to list the protocols in use, the local address and port numbers, the foreign address and port numbers, and the connection state.

```
C:\> netstat

Active Connections

  Proto Local Address          Foreign Address            State
  TCP   192.168.1.124:3126     192.168.0.2:netbios-ssn    ESTABLISHED
  TCP   192.168.1.124:3158     207.138.126.152:http       ESTABLISHED
  TCP   192.168.1.124:3159     207.138.126.169:http       ESTABLISHED
  TCP   192.168.1.124:3161     sc.msn.com:http            ESTABLISHED
  TCP   192.168.1.124:3166     www.cisco.com:http         ESTABLISHED
(output omitted)
C:\> netstat
```

The netstat Command (Cont.)

- By default, the **netstat** command will attempt to resolve IP addresses to domain names and port numbers to well-known applications. The **-n** option can display IP addresses and port numbers in numerical form.

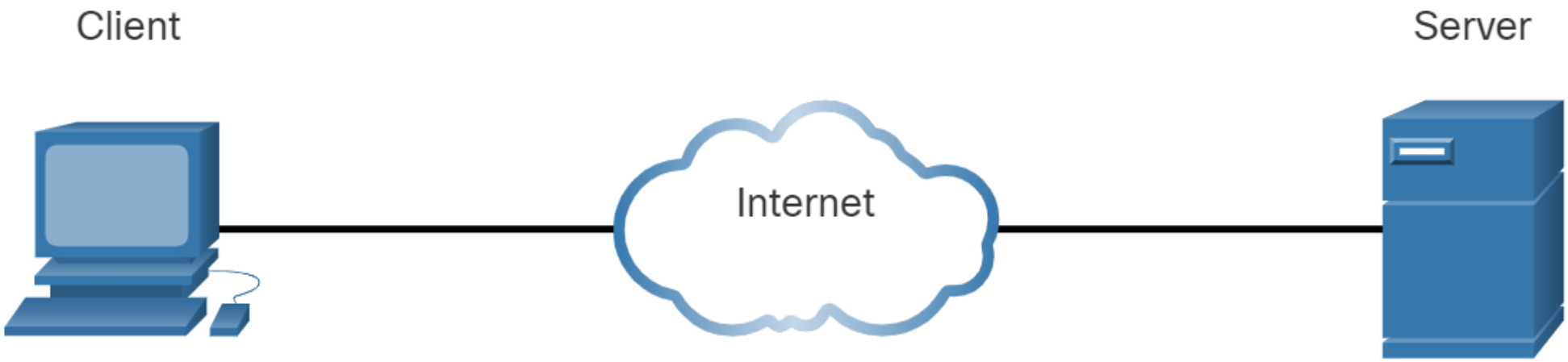
14. The Client Server Relationship

Client and Server Interaction

- Every day, we use the services available over networks and the internet to communicate with others and perform routine tasks. We rarely think of the servers, clients, and networking devices necessary to receive an email, update our status on social media, or shop for the best bargains in an online store. Most commonly used internet applications rely on complicated interactions between servers and clients.

- The term server refers to a host running a software application that provides information or services to other hosts connected to the network. A well-known example of an application is a web server. Millions of servers are connected to the internet, providing services such as websites, email, financial transactions, music downloads, etc. A crucial factor in enabling these complex interactions to function is that they all use agreed-upon standards and protocols.

Client and Server Interaction (Cont.)



Client and Server Interaction

An example of client software is a web browser like Chrome or Firefox. A single computer can also run multiple types of client software. For example, a user can check email and view a web page while instant messaging and listening to an audio stream. The table lists three common types of server software.

Type	Description
Email	The email server runs email server software. Clients use mail client software, such as Microsoft Outlook, to access email on the server.
Web	The web server runs web server software. Clients use browser software, such as Chrome or Firefox, to access web pages on the server.
File	The file server stores corporate and user files in a central location. The client devices access these files with client software such as Windows File Explorer.

URI, URN, and URL

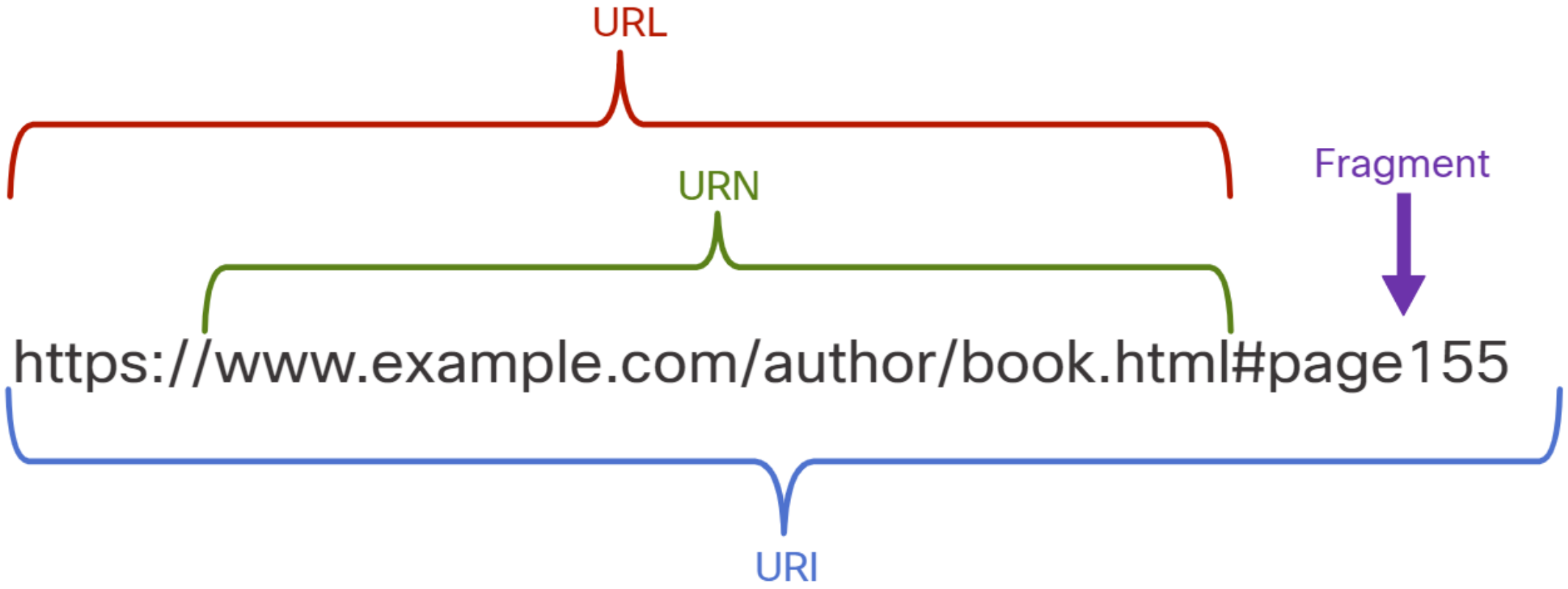
Web resources and web services such as RESTful APIs identify using a Uniform Resource Identifier (URI). A URI is a string of characters that identifies a specific network resource. As shown in the figure, a URI has two specializations:

- **Uniform Resource Name (URN)** - This identifies only the namespace of the resource (web page, document, image, etc.) without reference to the protocol.
- **Uniform Resource Locator (URL)** - This defines the network location of a specific resource on the network. Web browsers use HTTP or HTTPS URLs. Use a URL for protocols such as FTP, SFTP, SSH, and others. A URL using SFTP might look like this: sftp://sftp.example.com.

These are the parts of a URI, as shown in the figure:

- **Protocol/scheme** - HTTPS or other protocols such as FTP, SFTP, mailto, and NNTP
- **Hostname** - w ww.example.com
- **Path and file name** - /author/book.html
- **Fragment** - #page155

URI, URN, and URL (Cont.)

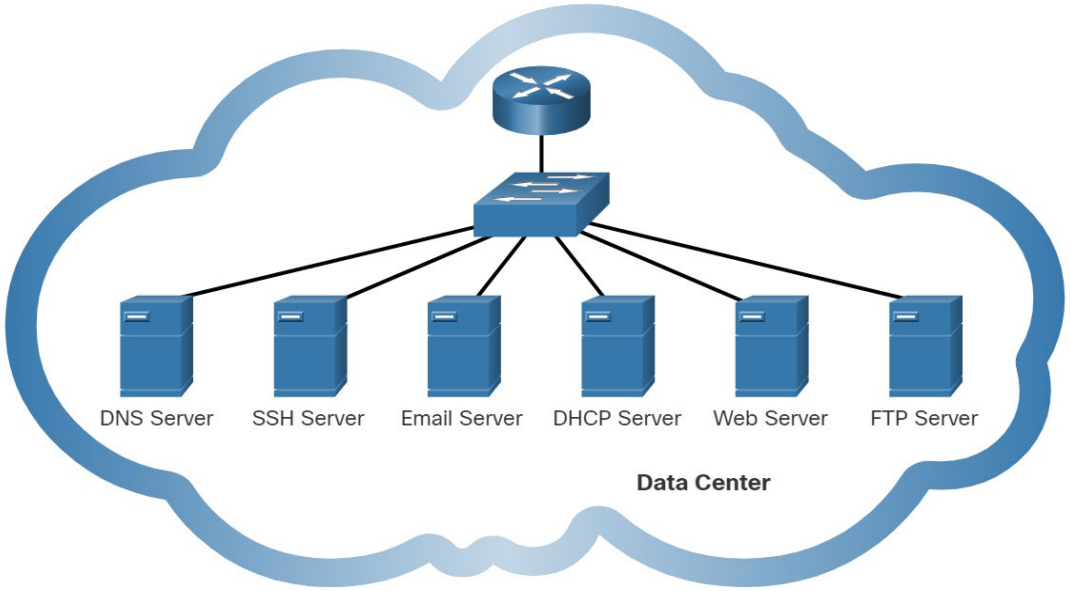


15. Network Application Services

Common Network Application Services

- What are the most common internet services that you use regularly? For most people, the list includes services such as internet searches, social media sites, video and audio streaming, online shopping sites, email, and messaging. Each of these services relies on protocols from the TCP/IP protocol suite to reliably communicate the information between the clients and the servers.

- The figure shows some of the most common servers that provide these services. The table shows a brief description of each service.



Common Network Application Services

Protocol	Description
Domain Name System (DNS)	Resolves internet names to IP addresses.
Secure Shell (SSH)	Used to provide remote access to servers and networking devices.
Simple Mail Transfer Protocol (SMTP)	Sends email messages and attachments from clients to servers and from servers to other email servers.
Post Office Protocol (POP)	Used by email clients to retrieve email and attachments from a remote server.
Internet Message Access Protocol (IMAP)	Used by email clients to retrieve email and attachments from a remote server.
Dynamic Host Configuration Protocol (DHCP)	Used to automatically configure devices with IP addressing and other necessary information to enable them to communicate over the internet.
Hypertext Transfer Protocol (HTTP)	Used by web browsers to request web pages and web servers to transfers the files that make up web pages of the World Wide Web.
File Transfer Protocol (FTP)	Used for interactive file transfer between systems.

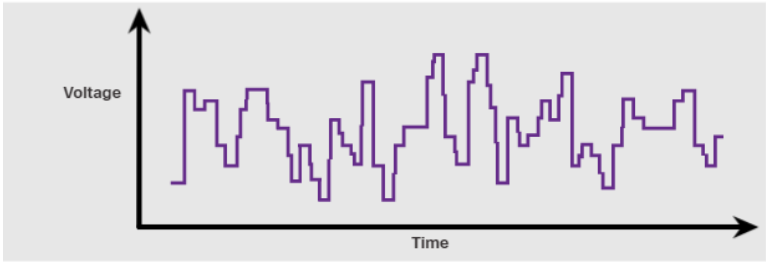
16. Domain Name System

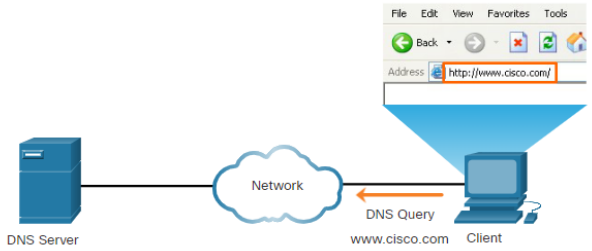
Domain Name System

- The design of other application layer-specific protocols is to make it easier to obtain addresses for network devices.
- They are essential because it would be time-consuming to remember IP addresses instead of URLs or manually configure all devices in a medium to a large network.
- In data networks, devices are labeled with numeric IP addresses to send and receive data.
- Domain names are created to convert the numeric address into a simple, recognizable name.
- If Cisco changes the numeric address of www.cisco.com, it is transparent to the user because the domain name remains the same.
- The new address links to the existing domain name and maintains connectivity.
- The DNS protocol defines an automated service that matches resource names with the required numeric network address, including the format for queries, responses, and data.
- The DNS protocol communications use a single format called a message. Its use is for all types of client queries and server responses, error messages, and the transfer of resource record information between servers.

Domain Name System (Cont.)

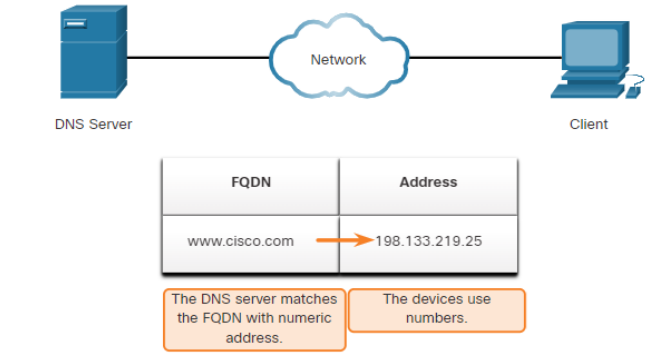
Step 1	The user types an FQDN into a browser application Address field.
Step 2	The designated DNS server for the client computer receives a DNS query.

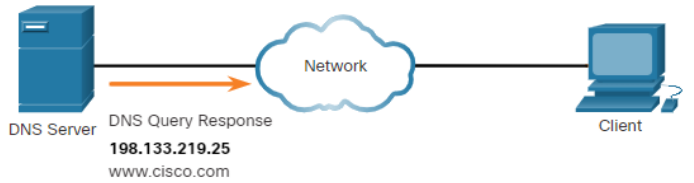




Domain Name System (Cont.)

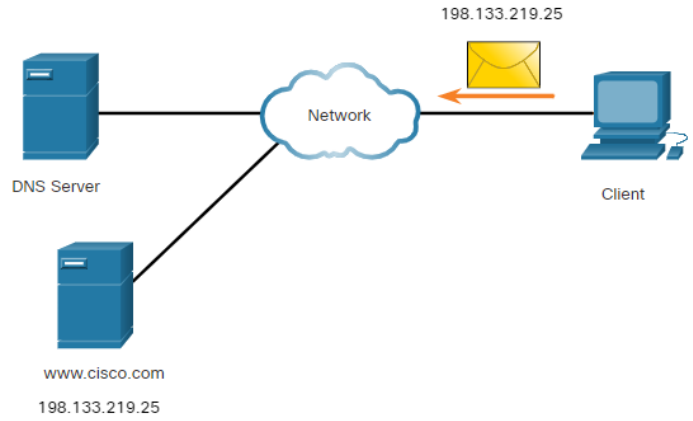
Step 3	The DNS server matches the FQDN with its IP address.
Step 4	The DNS query response is sent back to the client with the IP address for the FQDN.





Domain Name System (Cont.)

Step 5	The client computer uses the IP address to make requests to the server.



DNS Message Format

- The DNS server stores different types of resource records to resolve names.
- These records contain the name, address, and type of record.
- Some of these record types are as follows:

  - **A** - An end device IPv4 address
  - **NS** - An authoritative name server
  - **AAAA** - An end device IPv6 address (pronounced quad-A)
  - **MX** - A mail exchange record

- When a client queries, the server DNS process looks at its records to resolve the name.
- If it cannot resolve the name using its stored records, it contacts other servers to resolve the name.
- After a match is found and returned to the original requesting server, the server temporarily stores the numbered address if the same name is requested again.
- The DNS client service on Windows PCs also stores previously resolved names in memory.
- The **ipconfig /displaydns** command displays all the cached DNS entries.

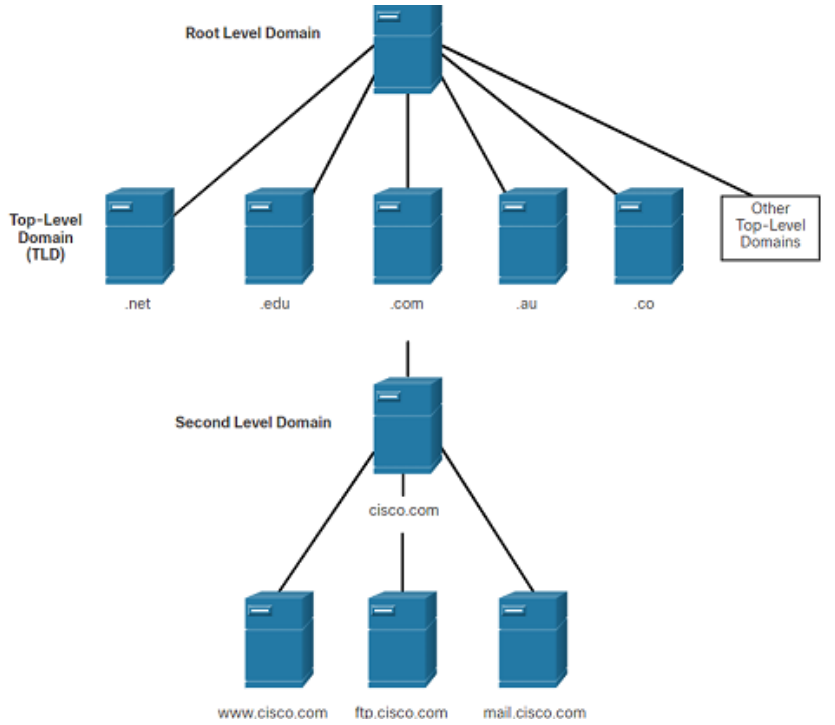
DNS Message Format (Cont.)

- As shown in the table, DNS uses the same message format between servers, consisting of a question, answer, authority, and additional information for all types of client queries and server responses, error messages, and transfer of resource record information.

Question	The question for the name server
Answer	Resource Records answering the question
Authority	Resource Records pointing toward an authority
Additional	Resource Records holding additional information

DNS Hierarchy

- The DNS protocol uses a hierarchical system to create a database to provide name resolution, as shown in the figure.
- DNS uses domain names to form the hierarchy.
- The naming structure breaks down into small, manageable zones.
- Each DNS server maintains a specific database file and is only responsible for managing name-to-IP mappings for that small portion of the entire DNS structure.
- When a DNS server receives a request for a name translation that is not within its DNS zone, the DNS server forwards the request to another DNS server within the proper zone for translation.
- DNS is scalable because hostname resolution spreads across multiple servers.
- The different top-level domains represent either the type of organization or the country of origin.
    - .com - a business or industry
    - .org - a non-profit organization
    - .au - Australia
    - .co - Colombia



The nslookup Command

- When configuring a network device, one or more DNS server addresses provide the DNS client for name resolution.
- Usually, the ISP provides the addresses to use for the DNS servers.
- When a user application requests to connect to a remote device by name, the requesting DNS client queries the name server to resolve the name to a numeric address.
- Computer operating systems also have a utility called nslookup that allows the user to manually query the name servers to resolve a given host name.
- Using this utility can also troubleshoot name resolution issues and verify the name servers' current status.
- In this figure, when the **nslookup** command is issued, the default DNS server configured for your host is displayed.
- The host or domain name can be entered at the **nslookup** prompt.

```
C:\Users> nslookup
Default Server:  dns-sj.cisco.com
Address:  171.70.168.183
> www.cisco.com
Server:  dns-sj.cisco.com
Address:  171.70.168.183
Name:    origin-www.cisco.com
Addresses:  2001:420:1101:1::a
          173.37.145.84
Aliases:  www.cisco.com
> cisco.netacad.net
Server:  dns-sj.cisco.com
Address:  171.70.168.183
Name:  cisco.netacad.net
Address:  72.163.6.223
>
```

17. Hypertext Transfer Protocol (HTTP)

HTTP and HTML

- When a web client receives the IP address of a web server, the client browser uses that IP address and port 80 to request web services. This request is sent to the server using the Hypertext Transfer Protocol (HTTP).

- When the server receives a port 80 request, the server responds to the client's request and sends the web page to the client. The information content of a web page is encoded using specialized 'mark-up' languages. The **HyperText Markup Language (HTML)** coding tells the browser how to format the web page and what graphics and fonts to use. HTML is the most commonly used language.

HTTP and HTML (Cont.)

- The HTTP protocol is not secure; other users could easily intercept information as data sends over the network. To provide security for the data, HTTP can be used with secure transport protocols. Requests for secure HTTP send to port 443. These requests use HTTPS in the site address in the browser rather than HTTP.

- There are many different web servers and web clients available. The HTTP protocol and HTML standards allow these servers and clients from many manufacturers to work together seamlessly.

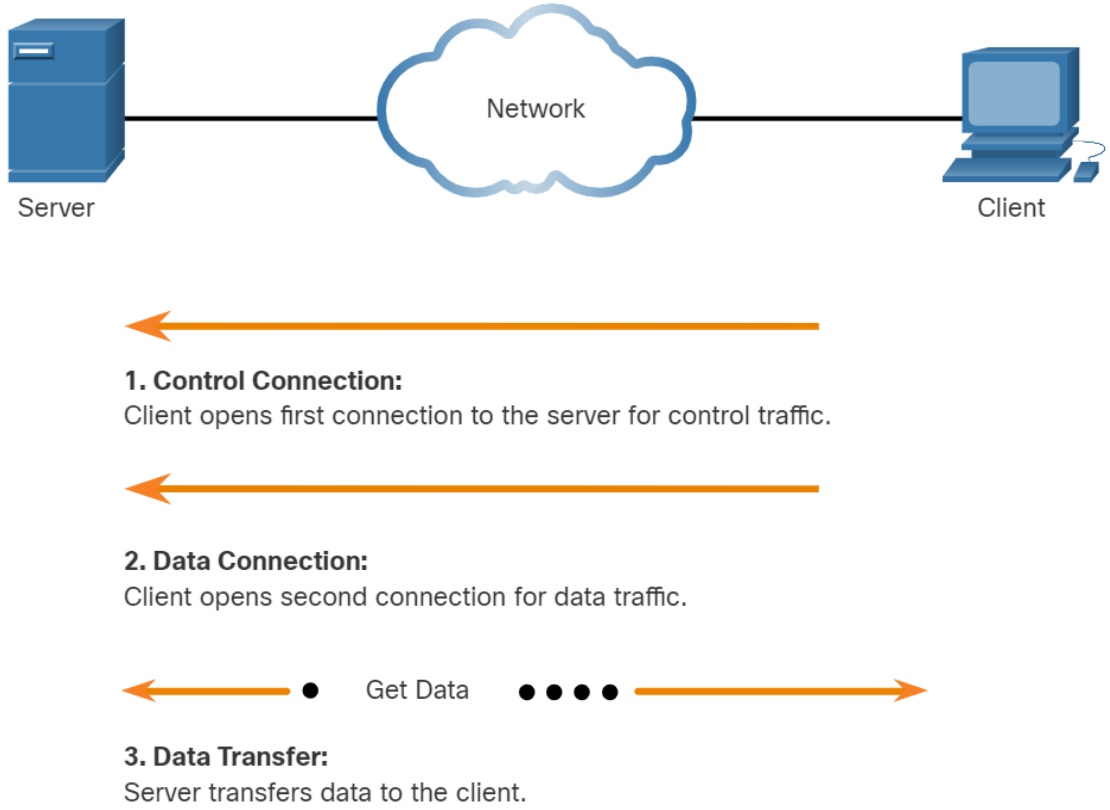
18. File Transfer Protocol (FTP)

File Transfer Protocol

- In addition to web services, another typical service used across the internet allows users to transfer files.
- The File Transfer Protocol (FTP) easily transfers files from one computer to another. An FTP client software host can access an FTP server to perform various file management functions, including file uploads and downloads.
- The FTP server enables a client to exchange files between devices. It also allows clients to manage files remotely by sending file management commands such as delete or rename. The FTP service uses two different ports to communicate between the client and the server.

File Transfer Protocol (Cont.)

- The example in the figure illustrates how FTP operates. Control connection requests are sent to the server to begin an FTP session using destination TCP port 21. When the session opens, the server uses TCP port 20 to transfer the data files.



File Transfer Protocol (Cont.)

- FTP client software builds into computer operating systems and most web browsers. Stand-alone FTP clients offer many options in an easy-to-use GUI-based interface.

- Based on commands sent across the control connection, data can be downloaded from the server or uploaded from the client.

19. Virtual Terminals

Telnet

- Before desktop computers with sophisticated graphical interfaces existed, people used text-based systems, often displaying terminals physically attached to a central computer. After networks became available, people needed a way to remotely access the computer systems as they did with the directly-attached terminals.

- Telnet develops to meet that need. Telnet dates back to the early 1970s and is among the oldest application layer protocols and services in the TCP/IP suite. Telnet provides a standard method of emulating text-based terminal devices over the data network. Telnet refers to the protocol and the client software that implements it. Telnet servers listen for client requests on TCP port 23.

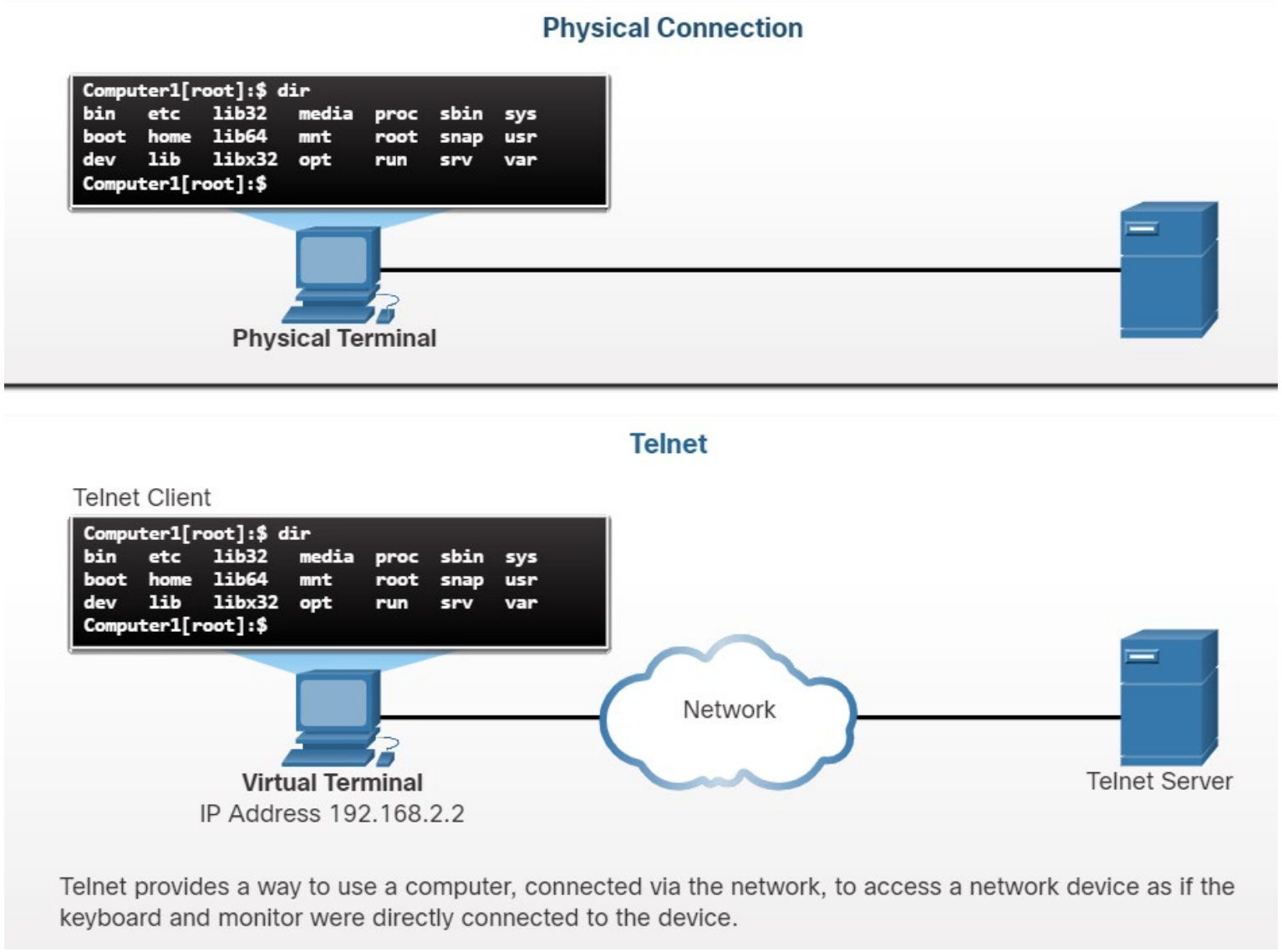
Telnet (Cont.)

- Appropriately enough, a Telnet connection is called a virtual terminal (vty) session or connection. Rather than using a physical device to connect to the server, Telnet uses software to create a virtual device that provides the same features of a terminal session with access to the server's command line interface (CLI).

- In the figure, the client has remotely connected to the server via Telnet. The client can now execute commands as if it locally connects to the server.

Note: Telnet does not consider to be a secure protocol. SSH should be used in most environments instead of Telnet. Telnet is used in several examples in this course for simplicity of configuration.

Telnet (Cont.)



Telnet provides a way to use a computer, connected via the network, to access a network device as if the keyboard and monitor were directly connected to the device.

Security Issues with Telnet

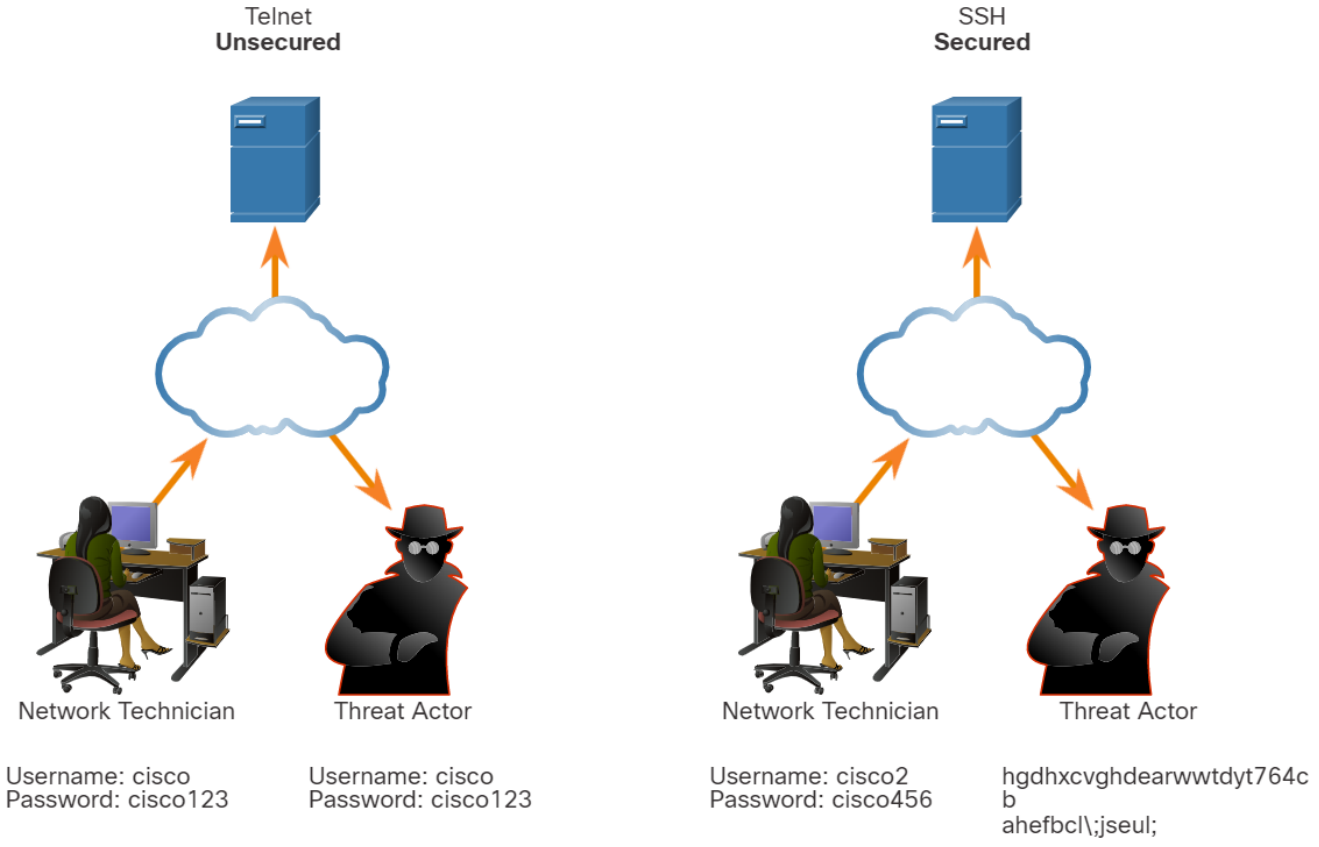
- After a Telnet connection establishes, users can perform any authorized function on the server, just as if they were using a command line session on the server itself. They can start and stop processes, configure the device, and even shut down the system. if authorized

- Although the Telnet protocol can require a user to log in, it does not support transporting encrypted data. Plaintext across the network transports as all data exchanges during Telnet sessions. This means that the data can be easily intercepted and understood.

- The **Secure Shell (SSH)** protocol offers an alternate and secure method for server access. SSH provides the structure for secure remote login and other secure network services. It also provides stronger authentication than Telnet and supports transporting session data using encryption. As a best practice, network professionals should always use SSH in place of Telnet whenever possible.

Security Issues with Telnet (Cont.)

- The figure illustrates how SSH is more secure than Telnet. Notice the use of Telnet when the data captured by the hacker is readable versus SSH when the data captured is encrypted and, therefore, more secure.

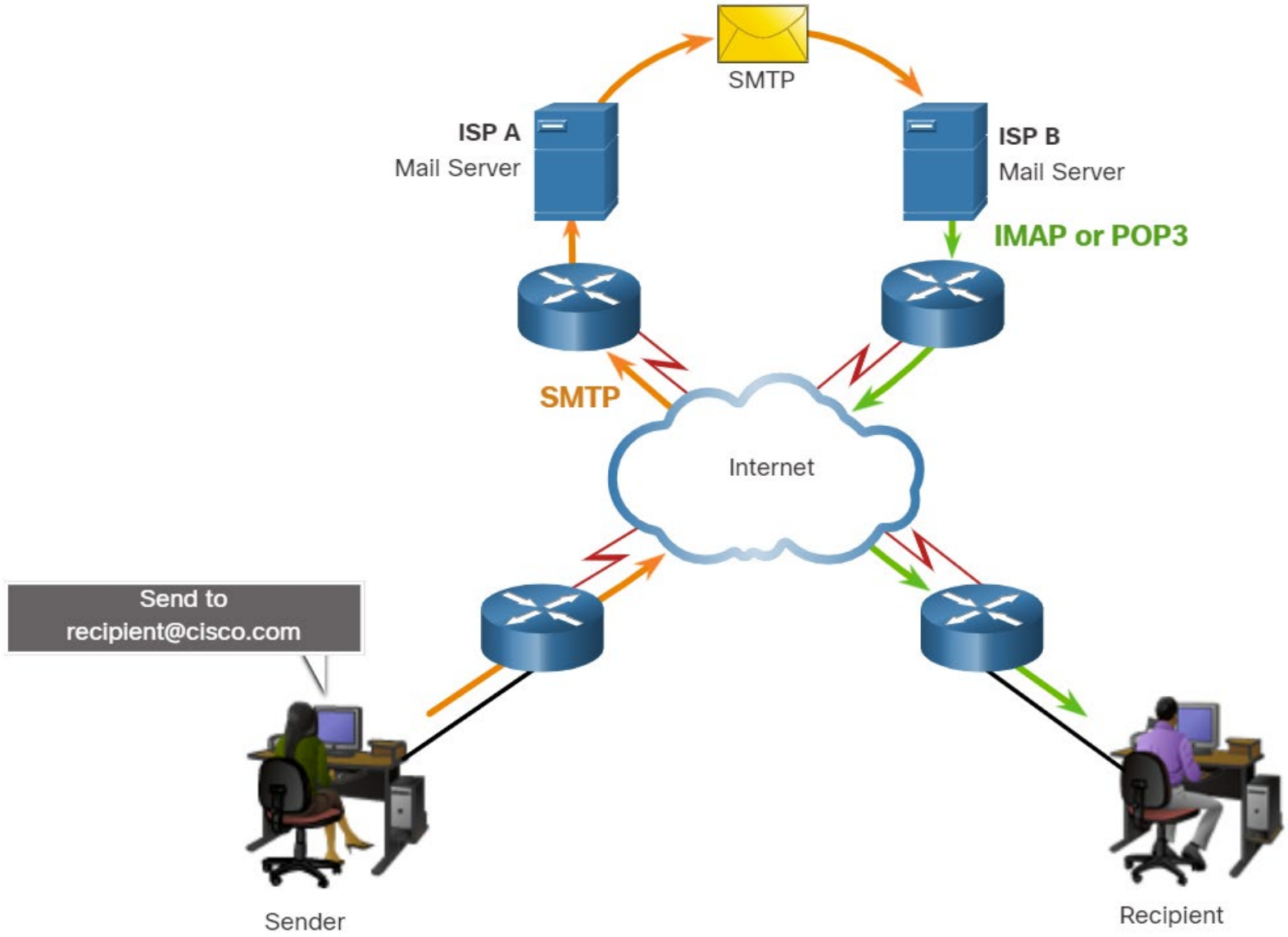


20. Email and Messaging

Email Clients and Servers

- Email is one of the internet's most popular clients/server applications. Email servers run server software, enabling them to interact with clients and other email servers over the network.
- Each mail server receives and stores mail for users who have mailboxes configured on the mail server. Each user with a mailbox must then use an email client to access the mail server and read these messages. Many internet messaging systems use a web-based client to access email. Examples of this type of client include Microsoft 365, Yahoo, and Gmail.
- This format identifies mailboxes: **user@company.domain**
- Various application protocols processing email include SMTP, POP3, and IMAP4.

Email Clients and Servers (Cont.)



Email Protocols

Simple Mail Transfer Protocol (SMTP)

- An email client uses SMTP to send messages to its local email server. The local server then decides if a local mailbox destines the message or if another server addresses the message.

- If the server has to send the message to a different server, it uses SMTP between those two servers. SMTP sends requests to port 25.

Email Protocols (Cont.)

Post Office Protocol (POP3)

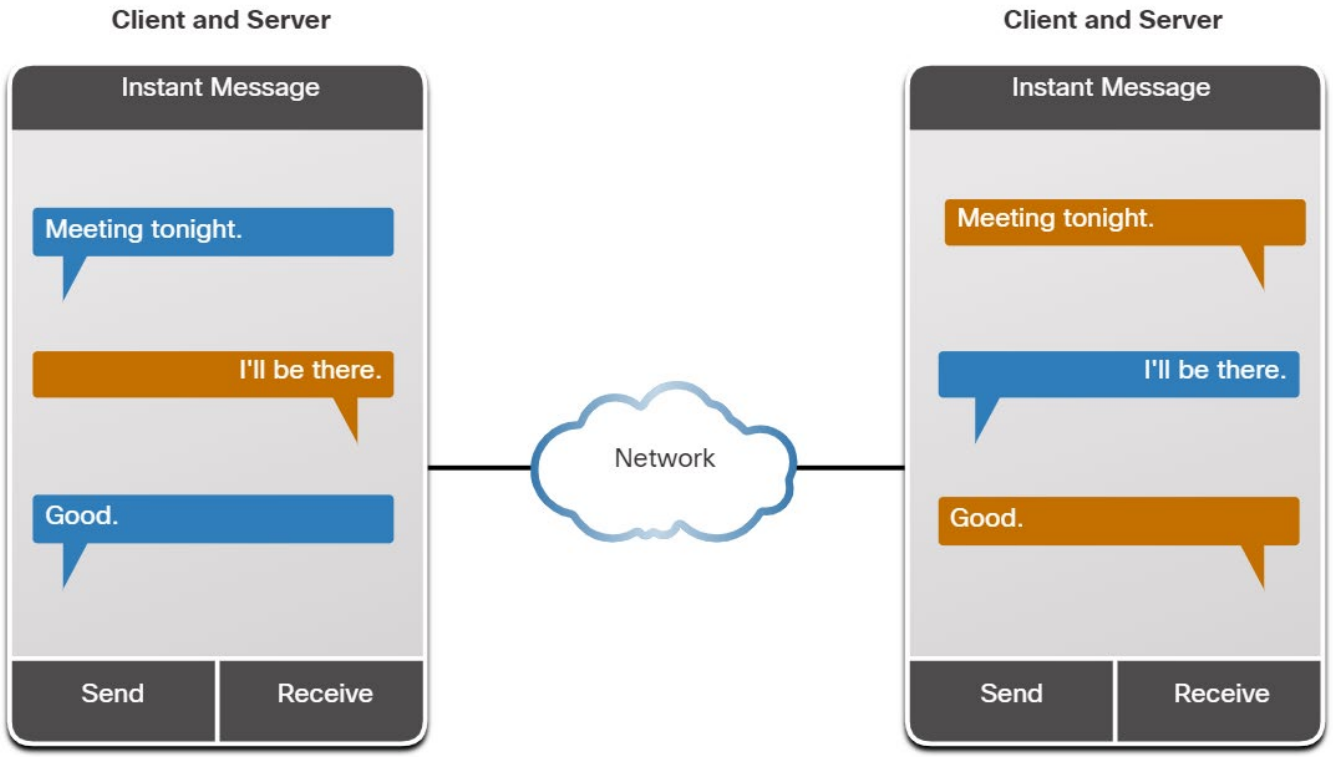
- A server that supports POP clients receives and stores messages addressed to its users. When the client connects to the email server, the messages download to the client. By default, messages are not kept on the server after the client has accessed them. Clients contact POP3 servers on port 110.

Internet Message Access Protocol (IMAP4)

- A server that supports IMAP clients also receives and stores messages addressed to its users. However, unlike POP, IMAP keeps the messages in the mailboxes on the server unless the user deletes them. The most current version of IMAP is IMAP4 which listens for client requests on port 143.
- Many different email servers exist for the various network operating system platforms.

Text Messaging

- Text messaging, shown in the figure, is one of the most popular communication tools today. In addition, text messaging software builds into many online applications, smartphone apps, and social media sites.



Text Messaging (Cont.)

- Both clients can simultaneously send and receive messages.
- Text messages may also be instant, direct, private, or chat. Text messaging enables users to communicate or chat over the internet in real time. A web-based client accesses text messaging services on a computer integrated into a social media or information-sharing site. These clients usually only connect to other users of the same site.
- There are also many standalone text message clients such as Cisco Webex Teams, Microsoft Teams, WhatsApp, Facebook Messenger, and others. These applications are available for various operating systems and devices and typically offer a mobile version. In addition to text messages, these clients support the transfer of documents, video, music, and audio files.

Internet Phone Calls

- Making telephone calls over the internet is becoming increasingly popular. As shown in the figure, an internet telephony client uses peer-to-peer technology similar to that used by instant messaging. IP telephony uses Voice over IP (VoIP) technology, which converts analog voice signals into digital data. The voice data encapsulates into IP packets that carry the phone call through the network.

- When the IP phone software installs, the user selects a unique name. This is so that other users can receive calls. Speakers and a microphone, built-in or separate, are required. A headset is frequently plugged into the computer to serve as a phone.

- Other users of the same service on the internet can make calls by selecting the username from a list. A call to a regular telephone (landline or cell phone) requires using a gateway to access the Public Switched Telephone Network (PSTN). Depending on the service, there may be charges associated with this type of call. The protocols and destination ports used by internet telephony applications can vary based on the software.

Internet Phone Calls (Cont.)

